

? When an earthquake strikes, the news of the event travels through the body of the earth in the form of seismic waves. Which aspects of a seismic wave determine how much power is carried by the wave: (i) the amplitude; (ii) the frequency; (iii) both the amplitude and the frequency; or (iv) neither the amplitude nor the frequency?



15 Mechanical Waves

LEARNING OUTCOMES

In this chapter, you'll learn...

- 15.1 What is meant by a mechanical wave, and the different varieties of mechanical waves.
- 15.2 How to use the relationship among speed, frequency, and wavelength for a periodic wave.
- 15.3 How to interpret and use the mathematical expression for a sinusoidal periodic wave.
- 15.4 How to calculate the speed of waves on a rope or string.
- 15.5 How to calculate the rate at which a mechanical wave transports energy.
- 15.6 What happens when mechanical waves overlap and interfere.
- 15.7 The properties of standing waves on a string, and how to analyze these waves.
- 15.8 How stringed instruments produce sounds of specific frequencies.

You'll need to review...

- 8.1 The impulse–momentum theorem.
- 14.1, 14.2 Periodic motion and simple harmonic motion.

Ripples on a pond, musical sounds, seismic tremors triggered by an earthquake—all these are *wave* phenomena. Waves can occur whenever a system is disturbed from equilibrium and when the disturbance can travel, or *propagate*, from one region of the system to another. As a wave propagates, it carries energy. The energy in light waves from the sun warms the surface of our planet; the energy in seismic waves can crack our planet's crust.

This chapter and the next are about mechanical waves—waves that travel within some material called a *medium*. (Chapter 16 is concerned with sound, an important type of mechanical wave.) We'll begin this chapter by deriving the basic equations for describing waves, including the important special case of *sinusoidal* waves in which the wave pattern is a repeating sine or cosine function. To help us understand waves in general, we'll look at the simple case of waves that travel on a stretched string or rope.

Waves on a string are important in music. When a musician strums a guitar or bows a violin, she makes waves that travel in opposite directions along the instrument's strings. What happens when these oppositely directed waves overlap is called *interference*. We'll discover that sinusoidal waves can occur on a guitar or violin string only for certain special frequencies, called *normal-mode frequencies*, determined by the properties of the string. These normal-mode frequencies determine the pitch of the musical sounds that a stringed instrument produces. (In the next chapter we'll find that interference also helps explain the pitches of *wind* instruments such as pipe organs.)

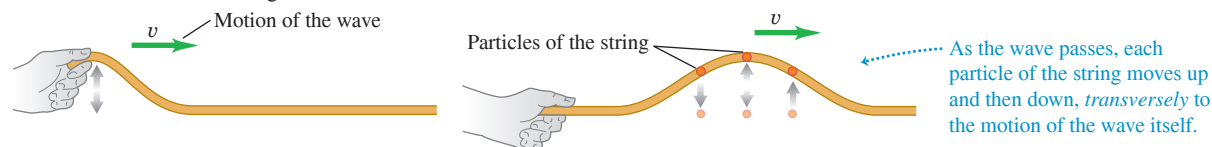
Not all waves are mechanical in nature. *Electromagnetic* waves—including light, radio waves, infrared and ultraviolet radiation, and x rays—can propagate even in empty space, where there is *no* medium. We'll explore these and other nonmechanical waves in later chapters.

15.1 TYPES OF MECHANICAL WAVES

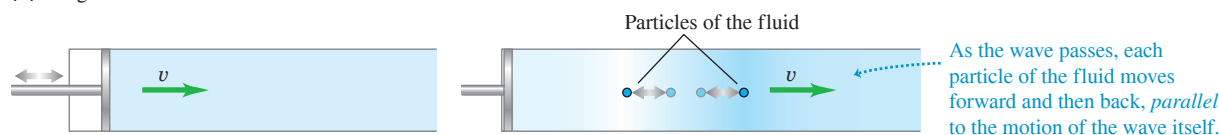
A **mechanical wave** is a disturbance that travels through some material or substance called the **medium** for the wave. As the wave travels through the medium, the particles that make up the medium undergo displacements of various kinds, depending on the nature of the wave.

Figure 15.1 Three ways to make a wave that moves to the right. (a) The hand moves the string up and then returns, producing a transverse wave. (b) The piston moves to the right, compressing the gas or liquid, and then returns, producing a longitudinal wave. (c) The board moves to the right and then returns, producing a combination of longitudinal and transverse waves.

(a) Transverse wave on a string



(b) Longitudinal wave in a fluid



(c) Waves on the surface of a liquid

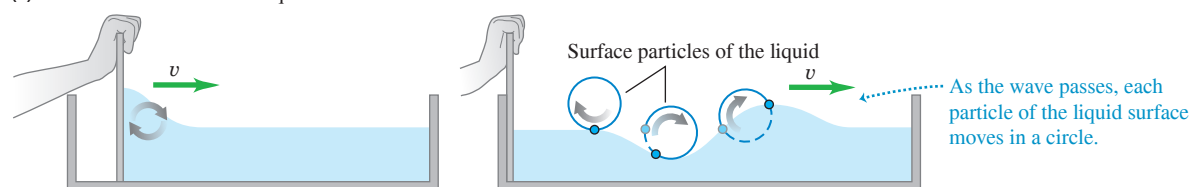


Figure 15.1 shows three varieties of mechanical waves. In Fig. 15.1a the medium is a string or rope under tension. If we give the left end a small upward shake or wiggle, the wiggle travels along the length of the string. Successive sections of string go through the same motion that we gave to the end, but at successively later times. Because the displacements of the medium are perpendicular or *transverse* to the direction of travel of the wave along the medium, this is called a **transverse wave**.

In Fig. 15.1b the medium is a liquid or gas in a tube with a rigid wall at the right end and a movable piston at the left end. If we give the piston a single back-and-forth motion, displacement and pressure fluctuations travel down the length of the medium. This time the motions of the particles of the medium are back and forth along the *same* direction that the wave travels. We call this a **longitudinal wave**.

In Fig. 15.1c the medium is a liquid in a channel, such as water in an irrigation ditch or canal. When we move the flat board at the left end forward and back once, a wave disturbance travels down the length of the channel. In this case the displacements of the water have *both* longitudinal and transverse components.

Each of these systems has an equilibrium state. For the stretched string it is the state in which the system is at rest, stretched out along a straight line. For the fluid in a tube it is a state in which the fluid is at rest with uniform pressure. And for the liquid in a trough it is a smooth, level water surface. In each case the wave motion is a disturbance from equilibrium that travels from one region of the medium to another. And in each case forces tend to restore the system to its equilibrium position when it is displaced, just as the force of gravity tends to pull a pendulum toward its straight-down equilibrium position when it is displaced.

These examples have three things in common. First, in each case the disturbance travels or *propagates* with a definite speed through the medium. This speed is called the speed of propagation, or simply the **wave speed**. Its value is determined in each case by the mechanical properties of the medium. We'll use the symbol v for wave speed. (The wave speed is *not* the same as the speed with which particles move when they are disturbed by the wave. We'll return to this point in Section 15.3.) Second, the medium itself does not travel through space; its individual particles undergo back-and-forth or up-and-down motions around their equilibrium positions. The overall pattern of the wave disturbance is what travels. Third, to set any of these systems into motion, we have to put in

BIO APPLICATION Waves on a Snake's Body

A snake moves itself along the ground by producing waves that travel backward along its body from its head to its tail. The waves remain stationary with respect to the ground as they push against the ground, so the snake moves forward.



Figure 15.2 “Doing the wave” at a sports stadium is an example of a mechanical wave: The disturbance propagates through the crowd, but there is no transport of matter (none of the spectators moves from one seat to another).



energy by doing mechanical work on the system. The wave motion transports this energy from one region of the medium to another. *Waves transport energy, but not matter, from one region to another (Fig. 15.2).*

TEST YOUR UNDERSTANDING OF SECTION 15.1 What type of wave is “the wave” shown in Fig. 15.2? (i) Transverse; (ii) longitudinal; (iii) a combination of transverse and longitudinal.

ANSWER (i) The “wave” travels horizontally from one spectator to the next along each row of the stadium, but the displacement of each spectator is vertically upward. Since the displacement is perpendicular to the direction in which the wave travels, the wave is transverse.

15.2 PERIODIC WAVES

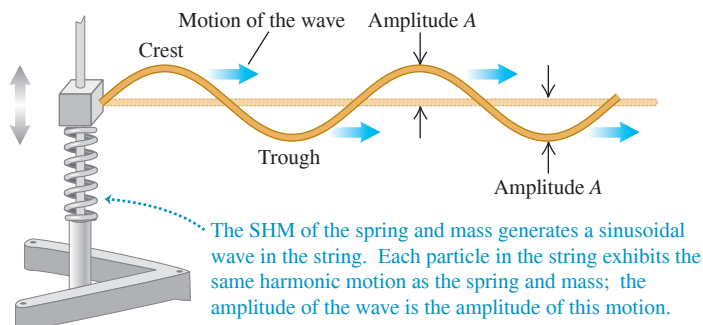
The transverse wave on a stretched string in Fig. 15.1a is an example of a *wave pulse*. The hand exerts a transverse force that shakes the string up and down just once, producing a single “wiggle,” or pulse, that travels along the length of the string. The tension in the string restores its straight-line shape once the pulse has passed.

A more interesting situation develops when we give the free end of the string a repetitive, or *periodic*, motion. (You should review the discussion of periodic motion in Chapter 14 before going ahead.) Each particle in the string undergoes periodic motion as the wave propagates, and we have a **periodic wave**.

Periodic Transverse Waves

Suppose we move one end of the string up and down with *simple harmonic motion* (SHM) as in Fig. 15.3, with amplitude A , frequency f , angular frequency $\omega = 2\pi f$, and period $T = 1/f = 2\pi/\omega$. The wave that results is a symmetric sequence of *crests* and *troughs*. As we’ll see, periodic waves with SHM are particularly easy to analyze; we

Figure 15.3 A block of mass m attached to a spring undergoes simple harmonic motion, producing a sinusoidal wave that travels to the right on the string. (In a real-life system a driving force would have to be applied to the block to replace the energy carried away by the wave.)



call them **sinusoidal waves**. It turns out that *any* periodic wave can be represented as a combination of sinusoidal waves. So this kind of wave motion is worth special attention.

In Fig. 15.3 the wave is a *continuous succession* of transverse sinusoidal disturbances. **Figure 15.4** shows the shape of a part of the string near the left end at time intervals of $\frac{1}{8}$ of a period, for a total time of one period. The wave shape advances steadily toward the right, as indicated by the highlighted area. As the wave moves, any point on the string (any of the red dots, for example) oscillates up and down about its equilibrium position with simple harmonic motion. *When a sinusoidal wave passes through a medium, every particle in the medium undergoes simple harmonic motion with the same frequency.*

CAUTION Wave motion vs. particle motion Don't confuse the motion of the *transverse wave* along the string and the motion of a *particle* of the string. The wave moves with constant speed v *along* the length of the string, while the motion of the particle is simple harmonic and *transverse* (perpendicular) to the length of the string. **|**

For a periodic wave, the shape of the string at any instant is a repeating pattern. The **wavelength** λ (the Greek letter lambda) of the wave is the distance from one crest to the next, or from one trough to the next, or from any point to the corresponding point on the next repetition of the wave shape. The wave pattern travels with constant speed v and advances a distance of one wavelength λ in a time interval of one period T . So the wave speed is $v = \lambda/T$ or, because $f = 1/T$ from Eq. (14.1),

$$\text{For a periodic wave: } v = \lambda f \quad (15.1)$$

Wave speed Wavelength
Frequency

The speed of propagation equals the product of wavelength and frequency. The frequency is a property of the *entire* periodic wave because all points on the string oscillate with the same frequency f .

Waves on a string propagate in just one dimension (in Fig. 15.4, along the x -axis). But the ideas of frequency, wavelength, and amplitude apply equally well to waves that propagate in two or three dimensions. **Figure 15.5** shows a wave propagating in two dimensions on the surface of a tank of water. As with waves on a string, the wavelength is the distance from one crest to the next, and the amplitude is the height of a crest above the equilibrium level.

In many important situations including waves on a string, the wave speed v is determined entirely by the mechanical properties of the medium. In this case, increasing f causes λ to decrease so the product $v = \lambda f$ remains the same, and waves of *all* frequencies propagate with the same wave speed. In this chapter we'll consider *only* waves of this kind. (In later chapters we'll study the propagation of light waves in transparent materials where the wave speed depends on frequency; this turns out to be the reason raindrops create a rainbow.)

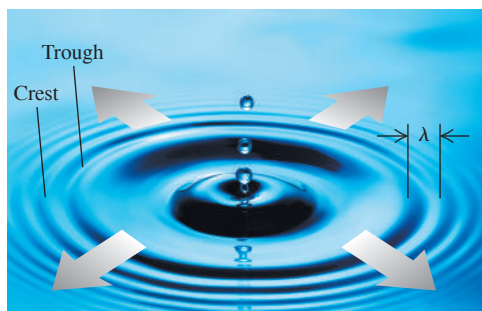


Figure 15.4 A sinusoidal transverse wave traveling to the right along a string. The vertical scale is exaggerated.

The string is shown at time intervals of $\frac{1}{8}$ period for a total of one period T . The highlighting shows the motion of one wavelength of the wave.

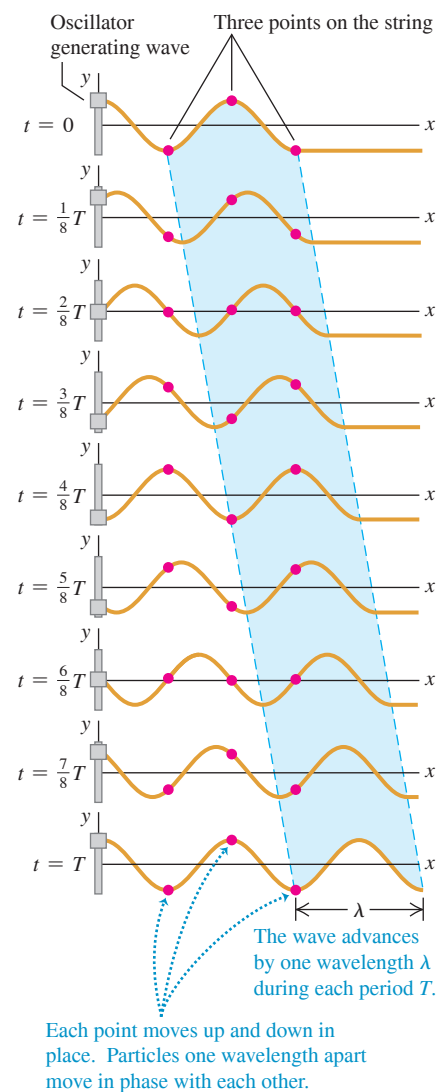
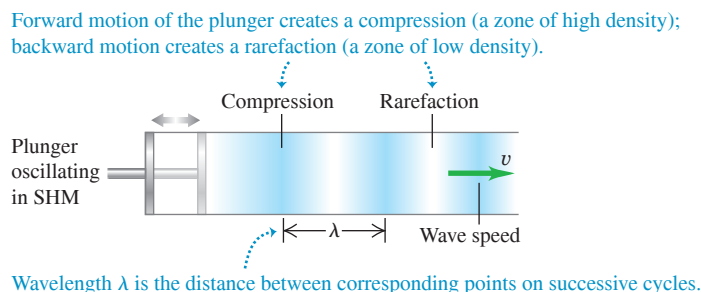
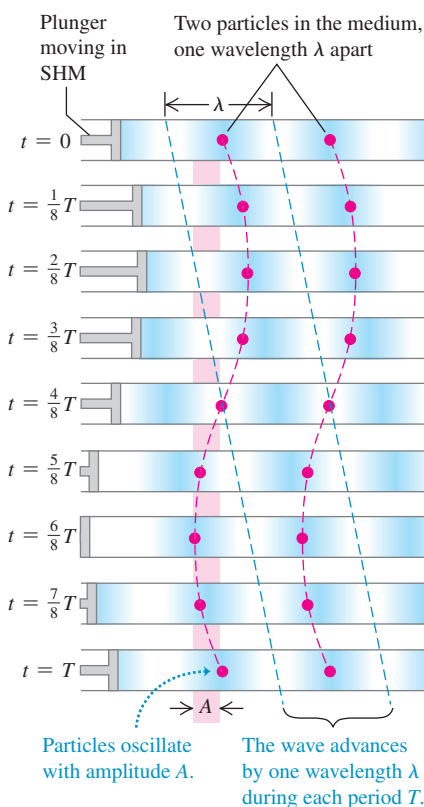


Figure 15.5 A series of drops falling into water produces a periodic wave that spreads radially outward. The wave crests and troughs are concentric circles. The wavelength λ is the radial distance between adjacent crests or adjacent troughs.

Figure 15.6 Using an oscillating piston to make a sinusoidal longitudinal wave in a fluid.

Figure 15.7 A sinusoidal longitudinal wave traveling to the right in a fluid. The wave has the same amplitude A and period T as the oscillation of the piston.

Longitudinal waves are shown at intervals of $\frac{1}{8}T$ for one period T .



Periodic Longitudinal Waves

To understand the mechanics of a periodic *longitudinal* wave, consider a long tube filled with a fluid, with a piston at the left end as in Fig. 15.1b. If we push the piston in, we compress the fluid near the piston, increasing the pressure in this region. This region then pushes against the neighboring region of fluid, and so on, and a wave pulse moves along the tube.

Now suppose we move the piston back and forth in SHM along a line parallel to the axis of the tube (Fig. 15.6). This motion forms regions in the fluid where the pressure and density are greater or less than the equilibrium values. We call a region of increased density a *compression*; a region of reduced density is a *rarefaction*. Figure 15.6 shows compressions as darkly shaded areas and rarefactions as lightly shaded areas. The wavelength is the distance from one compression to the next or from one rarefaction to the next.

Figure 15.7 shows the wave propagating in the fluid-filled tube at time intervals of $\frac{1}{8}$ of a period, for a total time of one period. The pattern of compressions and rarefactions moves steadily to the right, just like the pattern of crests and troughs in a sinusoidal transverse wave (compare Fig. 15.4). Each particle in the fluid oscillates in SHM parallel to the direction of wave propagation (that is, left and right) with the same amplitude A and period T as the piston. The particles shown by the two red dots in Fig. 15.7 are one wavelength apart, and so oscillate in phase with each other.

Just like the sinusoidal transverse wave shown in Fig. 15.4, in one period T the longitudinal wave in Fig. 15.7 travels one wavelength λ to the right. Hence the fundamental equation $v = \lambda f$ holds for longitudinal waves as well as for transverse waves, and indeed for *all* types of periodic waves. Just as for transverse waves, in this chapter and the next we'll consider only situations in which the speed of longitudinal waves does not depend on the frequency.

EXAMPLE 15.1 Wavelength of a musical sound

WITH VARIATION PROBLEMS

Sound waves are longitudinal waves in air. The speed of sound depends on temperature; at 20°C it is 344 m/s (1130 ft/s). What is the wavelength of a sound wave in air at 20°C if the frequency is 262 Hz (the approximate frequency of middle C on a piano)?

IDENTIFY and SET UP This problem involves Eq. (15.1), $v = \lambda f$, which relates wave speed v , wavelength λ , and frequency f for a periodic wave. The target variable is the wavelength λ . We are given $v = 344$ m/s and $f = 262$ Hz = 262 s⁻¹.

EXECUTE We solve Eq. (15.1) for λ :

$$\lambda = \frac{v}{f} = \frac{344 \text{ m/s}}{262 \text{ Hz}} = \frac{344 \text{ m/s}}{262 \text{ s}^{-1}} = 1.31 \text{ m}$$

EVALUATE The speed v of sound waves does *not* depend on the frequency. Hence $\lambda = v/f$ says that wavelength changes in inverse proportion to frequency. As an example, high (soprano) C is two octaves above middle C. Each octave corresponds to a factor of 2 in frequency, so the frequency of high C is four times that of middle C: $f = 4(262 \text{ Hz}) = 1048$ Hz. Hence the *wavelength* of high C is *one-fourth* as large: $\lambda = (1.31 \text{ m})/4 = 0.328 \text{ m}$.

KEYCONCEPT The product of a wave's wavelength and frequency has the same value no matter what the frequency is. This product equals the wave speed.

TEST YOUR UNDERSTANDING OF SECTION 15.2 If you double the wavelength of a wave on a particular string without changing the physical properties of the string, what happens to the wave speed v and the frequency f ? (i) v doubles and f is unchanged; (ii) v is unchanged and f doubles; (iii) v becomes one-half as great and f is unchanged; (iv) v is unchanged and f becomes one-half as great; (v) none of these.

ANSWER

(iv) The speed of waves on a string, v , does not depend on the wavelength. We can rewrite the relationship $v = \lambda f$ as $f = v/\lambda$, which tells us that if the wavelength λ doubles, the frequency f becomes one-half as great.

15.3 MATHEMATICAL DESCRIPTION OF A WAVE

Many characteristics of periodic waves can be described by using the concepts of wave speed, amplitude, period, frequency, and wavelength. Often, though, we need a more detailed description of the positions and motions of individual particles of the medium at particular times during wave propagation.

As a specific example, let's look at waves on a stretched string. If we ignore the sag of the string due to gravity, the equilibrium position of the string is along a straight line. We take this to be the x -axis of a coordinate system. Waves on a string are *transverse*; during wave motion a particle with equilibrium position x is displaced some distance y in the direction perpendicular to the x -axis. The value of y depends on which particle we are talking about (that is, y depends on x) and also on the time t when we look at it. Thus y is a *function* of both x and t ; $y = y(x, t)$. We call $y(x, t)$ the **wave function** that describes the wave. If we know this function for a particular wave motion, we can use it to find the displacement (from equilibrium) of any particle at any time. From this we can find the velocity and acceleration of any particle, the shape of the string, and anything else we want to know about the behavior of the string at any time.

Wave Function for a Sinusoidal Wave

Let's see how to determine the form of the wave function for a sinusoidal wave. Suppose a sinusoidal wave travels from left to right (the direction of increasing x) along the string, as in Fig. 15.8. Every particle of the string oscillates in simple harmonic motion with the same amplitude and frequency. But the oscillations of particles at different points on the string are *not* all in step with each other. The particle at point B in Fig. 15.8 is at its maximum positive value of y at $t = 0$ and returns to $y = 0$ at $t = \frac{2}{8}T$; these same events occur for a particle at point A or point C at $t = \frac{4}{8}T$ and $t = \frac{6}{8}T$, exactly one half-period later. For any two particles of the string, the motion of the particle on the right (in terms of the wave, the “downstream” particle) lags behind the motion of the particle on the left by an amount proportional to the distance between the particles.

Hence the cyclic motions of various points on the string are out of step with each other by various fractions of a cycle. We call these differences *phase differences*, and we say that the *phase* of the motion is different for different points. For example, if one point has its maximum positive displacement at the same time that another has its maximum negative displacement, the two are a half-cycle out of phase. (This is the case for points A and B, or points B and C.)

Suppose that the displacement of a particle at the left end of the string ($x = 0$), where the wave originates, is given by

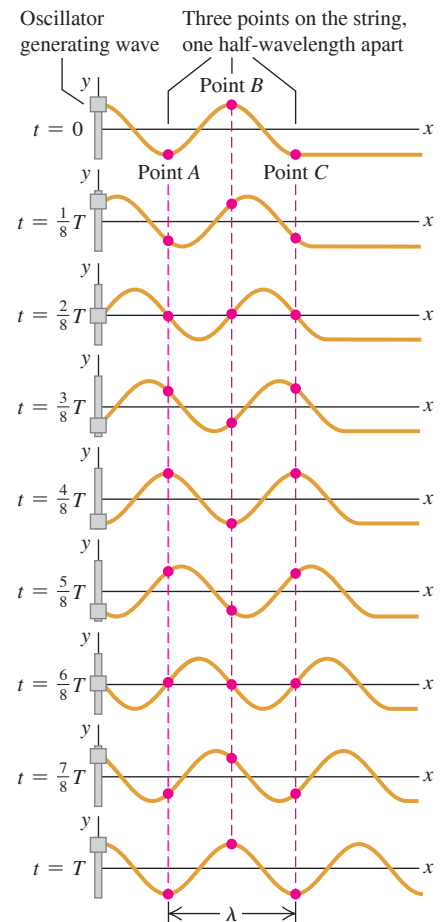
$$y(x = 0, t) = A \cos \omega t = A \cos 2\pi f t \quad (15.2)$$

That is, the particle oscillates in SHM with amplitude A , frequency f , and angular frequency $\omega = 2\pi f$. The notation $y(x = 0, t)$ reminds us that the motion of this particle is a special case of the wave function $y(x, t)$ that describes the entire wave. At $t = 0$ the particle at $x = 0$ is at its maximum positive displacement ($y = A$) and is instantaneously at rest (because y is a maximum).

The wave disturbance travels from $x = 0$ to some point x to the right of the origin in an amount of time given by x/v , where v is the wave speed. So the motion of point x at time t

Figure 15.8 Tracking the oscillations of three points on a string as a sinusoidal wave propagates along it.

The string is shown at time intervals of $\frac{1}{8}$ period for a total of one period T .



is the same as the motion of point $x = 0$ at the earlier time $t - x/v$. Hence we can find the displacement of point x at time t by simply replacing t in Eq. (15.2) by $(t - x/v)$:

$$y(x, t) = A \cos \left[\omega \left(t - \frac{x}{v} \right) \right]$$

Because $\cos(-\theta) = \cos \theta$, we can rewrite the wave function as

$$y(x, t) = A \cos \left[\omega \left(\frac{x}{v} - t \right) \right] \quad (15.3)$$

Wave function for a sinusoidal wave propagating in +x-direction

Amplitude

Position

Time

Angular frequency = $2\pi f$

Wave speed

The displacement $y(x, t)$ is a function of both the location x of the point and the time t . We could make Eq. (15.3) more general by allowing for different values of the phase angle, as we did for SHM in Section 14.2, but for now we omit this.

We can rewrite the wave function given by Eq. (15.3) in several different but useful forms. We can express it in terms of the period $T = 1/f$ and the wavelength $\lambda = v/f = 2\pi v/\omega$:

$$y(x, t) = A \cos \left[2\pi \left(\frac{x}{\lambda} - \frac{t}{T} \right) \right] \quad (15.4)$$

Wave function for a sinusoidal wave propagating in +x-direction

Amplitude

Position

Time

Wavelength

Period

It's convenient to define a quantity k , called the **wave number**:

$$k = \frac{2\pi}{\lambda} \quad (\text{wave number}) \quad (15.5)$$

Substituting $\lambda = 2\pi/k$ and $f = \omega/2\pi$ into Eq. (15.1), $v = \lambda f$, gives

$$\omega = vk \quad (\text{periodic wave}) \quad (15.6)$$

We can then rewrite Eq. (15.4) as

$$y(x, t) = A \cos(kx - \omega t) \quad (15.7)$$

Wave function for a sinusoidal wave propagating in +x-direction

Amplitude

Position

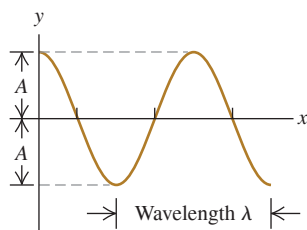
Time

Wave number = $2\pi/\lambda$

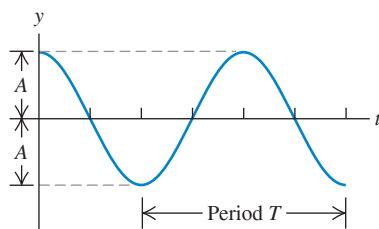
Angular frequency = $2\pi f$

Figure 15.9 Two graphs of the wave function $y(x, t)$ in Eq. (15.7). (a) Graph of displacement y versus coordinate x at time $t = 0$. (b) Graph of displacement y versus time t at coordinate $x = 0$. The vertical scale is exaggerated in both (a) and (b).

(a) If we use Eq. (15.7) to plot y as a function of x for time $t = 0$, the curve shows the *shape* of the string at $t = 0$.



(b) If we use Eq. (15.7) to plot y as a function of t for position $x = 0$, the curve shows the *displacement* y of the particle at $x = 0$ as a function of time.



Which of these various forms for the wave function $y(x, t)$ we use in any specific problem is a matter of convenience. Note that ω has units rad/s, so for unit consistency in Eqs. (15.6) and (15.7) the wave number k must have the units rad/m. (Warning: Some textbooks define the wave number as $1/\lambda$ rather than $2\pi/\lambda$.)

CAUTION Amplitude is independent of wavelength or frequency We learned in Section 15.2 that the frequency and wavelength of a wave are closely related: Increasing the frequency decreases the wavelength, and decreasing the frequency increases the wavelength. The amplitude of a wave, however, does not depend on the frequency or the wavelength. Changing the frequency or wavelength has no effect on the amplitude, and changing the wave amplitude by itself has no effect on the frequency or wavelength. ■

Graphing the Wave Function

Figure 15.9a graphs the wave function $y(x, t)$ as a function of x for a specific time t . This graph gives the displacement y of a particle from its equilibrium position as a function of the coordinate x of the particle. If the wave is a transverse wave on a string, the graph in Fig. 15.9a represents the shape of the string at that instant, like a flash photograph of the string. In particular, at time $t = 0$,

$$y(x, t = 0) = A \cos kx = A \cos 2\pi \frac{x}{\lambda}$$

Figure 15.9b is a graph of the wave function versus time t for a specific coordinate x . This graph gives the displacement y of the particle at x as a function of time; that is, it describes the motion of that particle. At position $x = 0$,

$$y(x = 0, t) = A \cos(-\omega t) = A \cos \omega t = A \cos 2\pi \frac{t}{T}$$

This is consistent with our original statement about the motion at $x = 0$, Eq. (15.2).

More on the Wave Function

We can modify Eqs. (15.3) through (15.7) to represent a wave traveling in the *negative* x -direction. In this case the displacement of point x at time t is the same as the motion of point $x = 0$ at the *later* time $(t + x/v)$, so in Eq. (15.2) we replace t by $(t + x/v)$. For a wave traveling in the negative x -direction,

$$y(x, t) = A \cos \left[\omega \left(\frac{x}{v} + t \right) \right] = A \cos \left[2\pi \left(\frac{x}{\lambda} + \frac{t}{T} \right) \right] = A \cos(kx + \omega t) \quad (15.8)$$

(sinusoidal wave moving in $-x$ -direction)

In the expression $y(x, t) = A \cos(kx \pm \omega t)$ for a wave traveling in the $-x$ - or $+x$ -direction, the quantity $(kx \pm \omega t)$ is called the **phase**. It plays the role of an angular quantity (always measured in radians) in Eq. (15.7) or (15.8), and its value for any values of x and t determines what part of the sinusoidal cycle is occurring at a particular point and time. For a crest (where $y = A$ and the cosine function has the value 1), the phase could be $0, \pm 2\pi, \pm 4\pi$, and so on; for a trough (where $y = -A$ and the cosine has the value -1), it could be $\pm \pi, \pm 3\pi, \pm 5\pi$, and so on.

The wave speed is the speed with which we have to move along with the wave to keep alongside a point of a given phase, such as a particular crest of a wave on a string. For a wave traveling in the $+x$ -direction, that means $kx - \omega t = \text{constant}$. Taking the derivative with respect to t , we find $k \, dx/dt = \omega$, or

$$\frac{dx}{dt} = \frac{\omega}{k}$$

Comparing this with Eq. (15.6), we see that dx/dt is equal to the speed v of the wave. Because of this relationship, v is sometimes called the *phase velocity* of the wave. (*Phase speed* would be a better term.)

CAUTION Wave graphs Although they may look the same, Figs. 15.9a and 15.9b are *not* identical. Figure 15.9a is a picture of the shape of the string at $t = 0$, while Fig. 15.9b is a graph of the displacement y of a particle at $x = 0$ as a function of time. 

PROBLEM-SOLVING STRATEGY 15.1 Mechanical Waves

IDENTIFY *the relevant concepts:* As always, identify the target variables; these may include mathematical *expressions* (for example, the wave function for a given situation). Note that wave problems fall into two categories. *Kinematics* problems, concerned with describing wave motion, involve wave speed v , wavelength λ (or wave number k), frequency f (or angular frequency ω), and amplitude A . They may also involve the position, velocity, and acceleration of individual particles in the medium. *Dynamics* problems also use concepts from Newton's laws. Later in this chapter we'll encounter problems that involve the relationship of wave speed to the mechanical properties of the medium.

SET UP *the problem* using the following steps:

1. List the given quantities. Sketch graphs of y versus x (like Fig. 15.9a) and of y versus t (like Fig. 15.9b), and label them with known values.
2. Identify useful equations. These may include Eq. (15.1) ($v = \lambda f$), Eq. (15.6) ($\omega = vk$), and Eqs. (15.3), (15.4), and (15.7), which

express the wave function in various forms. From the wave function, you can find the value of y at any point (value of x) and at any time t .

3. If you need to determine the wave speed v and don't know both λ and f , you may be able to use a relationship between v and the mechanical properties of the system. (In the next section we'll develop this relationship for waves on a string.)

EXECUTE *the solution:* Solve for the unknown quantities using the equations you've identified. To determine the wave function from Eq. (15.3), (15.4), or (15.7), you must know A and any two of v , λ , and f (or v , k , and ω).

EVALUATE *your answer:* Confirm that the values of v , f , and λ (or v , ω , and k) agree with the relationships given in Eq. (15.1) or (15.6). If you've calculated the wave function, check one or more special cases for which you can predict the results.

EXAMPLE 15.2 Wave on a clothesline

WITH VARIATION PROBLEMS

Cousin Throckmorton holds one end of the clothesline taut and wiggles it up and down sinusoidally with frequency 2.00 Hz and amplitude 0.075 m. The wave speed on the clothesline is $v = 12.0$ m/s. At $t = 0$ Throcky's end has maximum positive displacement and is instantaneously at rest. Assume that no wave bounces back from the far end. (a) Find the wave amplitude A , angular frequency ω , period T , wavelength λ , and wave number k . (b) Write a wave function describing the wave. (c) Write equations for the displacement, as a function of time, of Throcky's end of the clothesline and of a point 3.00 m from that end.

IDENTIFY and SET UP This is a kinematics problem about the clothesline's wave motion. Throcky produces a sinusoidal wave that propagates along the clothesline, so we can use all of the expressions of this section. In part (a) our target variables are A , ω , T , λ , and k . We use the relationships $\omega = 2\pi f$, $f = 1/T$, $v = \lambda f$, and $k = 2\pi/\lambda$. In parts (b) and (c) our target "variables" are expressions for displacement, which we'll obtain from an appropriate equation for the wave function. We take the positive x -direction to be the direction in which the wave propagates, so either Eq. (15.4) or (15.7) will yield the desired expression. A photograph of the clothesline at time $t = 0$ would look like Fig. 15.9a, with the maximum displacement at $x = 0$ (the end that Throcky holds).

EXECUTE (a) The wave amplitude and frequency are the same as for the oscillations of Throcky's end of the clothesline, $A = 0.075$ m and $f = 2.00$ Hz. Hence

$$\begin{aligned}\omega &= 2\pi f = \left(2\pi \frac{\text{rad}}{\text{cycle}}\right) \left(2.00 \frac{\text{cycles}}{\text{s}}\right) \\ &= 4.00\pi \text{ rad/s} = 12.6 \text{ rad/s}\end{aligned}$$

The period is $T = 1/f = 0.500$ s, and from Eq. (15.1),

$$\lambda = \frac{v}{f} = \frac{12.0 \text{ m/s}}{2.00 \text{ s}^{-1}} = 6.00 \text{ m}$$

We find the wave number from Eq. (15.5) or (15.6):

$$k = \frac{2\pi}{\lambda} = \frac{2\pi \text{ rad}}{6.00 \text{ m}} = 1.05 \text{ rad/m}$$

or

$$k = \frac{\omega}{v} = \frac{4.00\pi \text{ rad/s}}{12.0 \text{ m/s}} = 1.05 \text{ rad/m}$$

(b) We write the wave function using Eq. (15.4) and the values of A , T , and λ from part (a):

$$\begin{aligned}y(x, t) &= A \cos 2\pi \left(\frac{x}{\lambda} - \frac{t}{T} \right) \\ &= (0.075 \text{ m}) \cos 2\pi \left(\frac{x}{6.00 \text{ m}} - \frac{t}{0.500 \text{ s}} \right) \\ &= (0.075 \text{ m}) \cos [(1.05 \text{ rad/m})x - (12.6 \text{ rad/s})t]\end{aligned}$$

We can also get this same expression from Eq. (15.7) by using the values of ω and k from part (a).

(c) We can find the displacement as a function of time at $x = 0$ and $x = +3.00$ m by substituting these values into the wave function from part (b):

$$\begin{aligned}y(x = 0, t) &= (0.075 \text{ m}) \cos 2\pi \left(\frac{0}{6.00 \text{ m}} - \frac{t}{0.500 \text{ s}} \right) \\ &= (0.075 \text{ m}) \cos (12.6 \text{ rad/s})t \\ y(x = +3.00 \text{ m}, t) &= (0.075 \text{ m}) \cos 2\pi \left(\frac{3.00 \text{ m}}{6.00 \text{ m}} - \frac{t}{0.500 \text{ s}} \right) \\ &= (0.075 \text{ m}) \cos [\pi - (12.6 \text{ rad/s})t] \\ &= -(0.075 \text{ m}) \cos (12.6 \text{ rad/s})t\end{aligned}$$

EVALUATE In part (b), the quantity $(1.05 \text{ rad/m})x - (12.6 \text{ rad/s})t$ is the *phase* of a point x on the string at time t . The two points in part (c) oscillate in SHM with the same frequency and amplitude, but their oscillations differ in phase by $(1.05 \text{ rad/m})(3.00 \text{ m}) = 3.15 \text{ rad} = \pi$ radians within rounding error—that is, one half-cycle—because the points are separated by one half-wavelength: $\lambda/2 = (6.00 \text{ m})/2 = 3.00 \text{ m}$. Thus, while a graph of y versus t for the point at $x = 0$ is a cosine curve (like Fig. 15.9b), a graph of y versus t for the point $x = 3.00$ m is a *negative* cosine curve (the same as a cosine curve shifted by one half-cycle).

Using the expression for $y(x = 0, t)$ in part (c), can you show that the end of the string at $x = 0$ is instantaneously at rest at $t = 0$, as stated at the beginning of this example? (*Hint:* Calculate the y -velocity at this point by taking the derivative of y with respect to t .)

KEYCONCEPT The wave function of a wave describes the displacement from equilibrium of the wave medium. It gives this displacement at any position in the medium and at any time.

Particle Velocity and Acceleration in a Sinusoidal Wave

From the wave function we can get an expression for the transverse velocity of any *particle* in a transverse wave. We call this v_y to distinguish it from the wave propagation speed v . To find the transverse velocity v_y at a particular point x , we take the derivative of the wave function $y(x, t)$ with respect to t , keeping x constant. If the wave function is

$$y(x, t) = A \cos(kx - \omega t)$$

then

$$v_y(x, t) = \frac{\partial y(x, t)}{\partial t} = \omega A \sin(kx - \omega t) \quad (15.9)$$

The ∂ in this expression is a modified d , used to remind us that $y(x, t)$ is a function of *two* variables and that we are allowing only one (t) to vary. The other (x) is constant because we are looking at a particular point on the string. This derivative is called a *partial*

derivative. If you haven't yet encountered partial derivatives in your study of calculus, don't fret; it's a simple idea.

Equation (15.9) shows that the transverse velocity of a particle varies with time, as we expect for simple harmonic motion. The maximum particle speed is ωA ; this can be greater than, less than, or equal to the wave speed v , depending on the amplitude and frequency of the wave.

The *acceleration* of any particle is the *second* partial derivative of $y(x, t)$ with respect to t :

$$\begin{aligned} a_y(x, t) &= \frac{\partial^2 y(x, t)}{\partial t^2} = -\omega^2 A \cos(kx - \omega t) \\ &= -\omega^2 y(x, t) \end{aligned} \quad (15.10)$$

The acceleration of a particle equals $-\omega^2$ times its displacement, which is the result we obtained in Section 14.2 for simple harmonic motion.

We can also compute partial derivatives of $y(x, t)$ with respect to x , holding t constant. The first partial derivative $\partial y(x, t)/\partial x$ is the *slope* of the string at point x and at time t . The second partial derivative with respect to x tells us the *curvature* of the string:

$$\frac{\partial^2 y(x, t)}{\partial x^2} = -k^2 A \cos(kx - \omega t) = -k^2 y(x, t) \quad (15.11)$$

From Eqs. (15.10) and (15.11) and the relationship $\omega = vk$ we see that

$$\frac{\partial^2 y(x, t)/\partial t^2}{\partial^2 y(x, t)/\partial x^2} = \frac{\omega^2}{k^2} = v^2 \quad \text{and}$$

Wave equation
involves second
partial derivatives
of wave function:

Second partial derivative with respect to x

$$\frac{\partial^2 y(x, t)}{\partial x^2}$$

Wave speed v^2

$= \frac{1}{v^2} \frac{\partial^2 y(x, t)}{\partial t^2}$

Second partial derivative
with respect to t

$$\frac{\partial^2 y(x, t)}{\partial t^2} \quad (15.12)$$

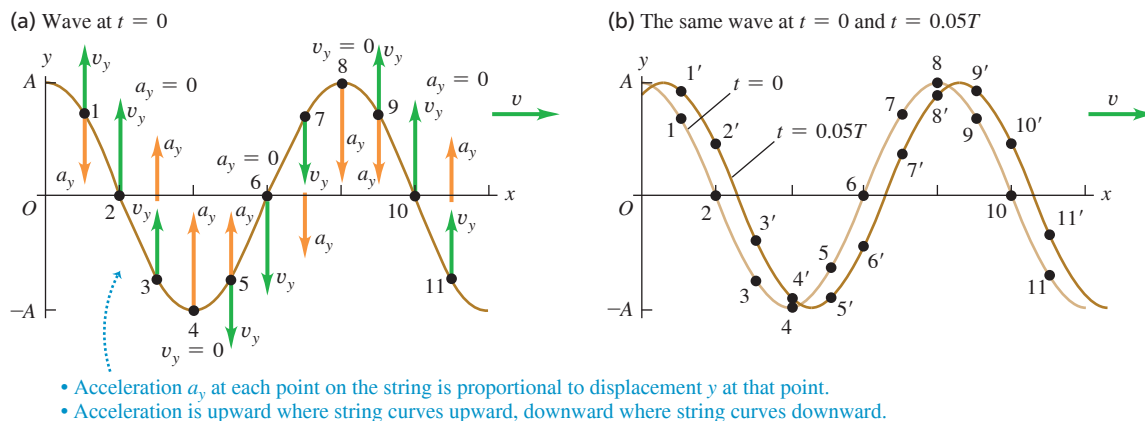
We've derived Eq. (15.12) for a wave traveling in the positive x -direction. You can show that the wave function for a sinusoidal wave propagating in the *negative* x -direction, $y(x, t) = A \cos(kx + \omega t)$, also satisfies this equation.

Equation (15.12), called the **wave equation**, is one of the most important equations in all of physics. Whenever it occurs, we know that a disturbance can propagate as a wave along the x -axis with wave speed v . The disturbance need not be a sinusoidal wave; we'll see in the next section that *any* wave on a string obeys Eq. (15.12), whether the wave is periodic or not. In Chapter 32 we'll find that electric and magnetic fields satisfy the wave equation; the wave speed will turn out to be the speed of light, which will lead us to the conclusion that light is an electromagnetic wave.

Figure 15.10a (next page) shows the transverse velocity v_y and transverse acceleration a_y , given by Eqs. (15.9) and (15.10), for several points on a string as a sinusoidal wave passes along it. At points where the string has an upward curvature ($\partial^2 y/\partial x^2 > 0$), the acceleration is positive ($a_y = \partial^2 y/\partial t^2 > 0$); this follows from the wave equation, Eq. (15.12). For the same reason the acceleration is negative ($a_y = \partial^2 y/\partial t^2 < 0$) at points where the string has a downward curvature ($\partial^2 y/\partial x^2 < 0$), and the acceleration is zero ($a_y = \partial^2 y/\partial t^2 = 0$) at points of inflection where the curvature is zero ($\partial^2 y/\partial x^2 = 0$). Remember that v_y and a_y are the *transverse* velocity and acceleration of points on the string; these points move along the y -direction, not along the propagation direction of the wave. Figure 15.10b shows these motions for several points on the string.

For *longitudinal* waves, the wave function $y(x, t)$ still measures the displacement of a particle of the medium from its equilibrium position. The difference is that for a longitudinal wave, this displacement is *parallel* to the x -axis instead of perpendicular to it. We'll discuss longitudinal waves in detail in Chapter 16.

Figure 15.10 (a) Another view of the wave at $t = 0$ in Fig. 15.9a. The vectors show the transverse velocity v_y and transverse acceleration a_y at several points on the string. (b) From $t = 0$ to $t = 0.05T$, a particle at point 1 is displaced to point 1', a particle at point 2 is displaced to point 2', and so on.



TEST YOUR UNDERSTANDING OF SECTION 15.3 Figure 15.8 shows a sinusoidal wave of period T on a string at times $0, \frac{1}{8}T, \frac{2}{8}T, \frac{3}{8}T, \frac{4}{8}T, \frac{5}{8}T, \frac{6}{8}T, \frac{7}{8}T$, and T . (a) At which time is point A on the string moving upward with maximum speed? (b) At which time does point B on the string have the greatest upward acceleration? (c) At which time does point C on the string have a downward acceleration and a downward velocity?

ANSWER

(a) $\frac{1}{8}T$. (b) $\frac{3}{8}T$. (c) $\frac{5}{8}T$. Since the wave is sinusoidal, each point on the string oscillates in simple harmonic motion (SHM). Hence we can apply all of the ideas from Chapter 14 about SHM to the wave depicted in Fig. 15.8. (a) A particle in SHM has its maximum speed when it is passing through the equilibrium position ($y = 0$ in Fig. 15.8). The particle at point A is moving upward through this position at $t = \frac{1}{8}T$. (b) In vertical SHM the greatest upward acceleration occurs when a particle is at its maximum downward displacement. This occurs for the particle at point B at $t = \frac{3}{8}T$. (c) A particle in vertical SHM has a downward acceleration when its displacement is upward. The particle at point C has an upward displacement and is moving downward at $t = \frac{5}{8}T$.

15.4 SPEED OF A TRANSVERSE WAVE

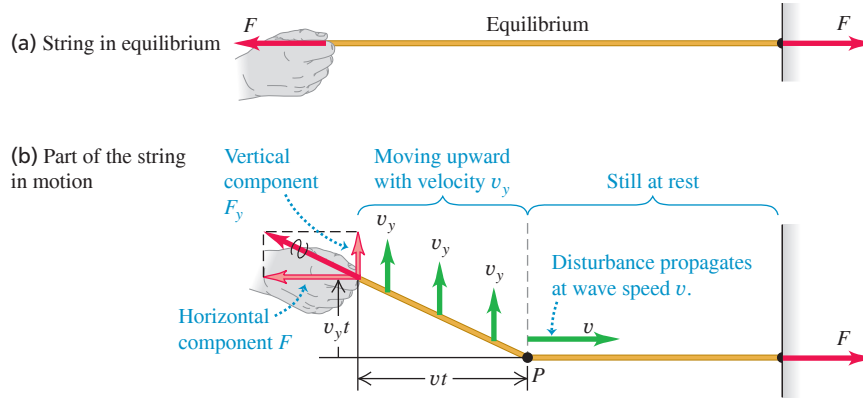
One of the key properties of any wave is the wave *speed*. Light waves in air have a much greater speed of propagation than do sound waves in air (3.00×10^8 m/s versus 344 m/s); that's why you see the flash from a bolt of lightning before you hear the clap of thunder. In this section we'll see what determines the speed of propagation of one particular kind of wave: transverse waves on a string. The speed of these waves is important to understand because it is an essential part of analyzing stringed musical instruments, as we'll discuss later in this chapter. Furthermore, the speeds of many kinds of mechanical waves turn out to have the same basic mathematical expression as does the speed of waves on a string.

What determines the speed of transverse waves on a string are the *tension* in the string and its *mass per unit length* (also called *linear mass density*). Increasing the tension also increases the restoring forces that tend to straighten the string when it is disturbed, thus increasing the wave speed. Increasing the mass per unit length makes the motion more sluggish, and so decreases the wave speed. We'll develop the exact relationship among wave speed, tension, and mass per unit length by two different methods. The first is simple in concept and considers a specific wave shape; the second is more general but also more formal.

Wave Speed on a String: First Method

We consider a perfectly flexible string (Fig. 15.11). In the equilibrium position the tension is F and the linear mass density (mass per unit length) is μ . (When portions of the string are displaced from equilibrium, the mass per unit length decreases a little,

Figure 15.11 Propagation of a transverse wave on a string.



and the tension increases a little.) We ignore the weight of the string so that when the string is at rest in the equilibrium position, the string forms a perfectly straight line as in Fig. 15.11a.

Starting at time $t = 0$, we apply a constant upward force F_y at the left end of the string. We might expect that the end would move with constant acceleration; that would happen if the force were applied to a *point* mass. But here the effect of the force F_y is to set successively more and more mass in motion. The wave travels with constant speed v , so the division point P between moving and nonmoving portions moves with the same constant speed v (Fig. 15.11b).

Figure 15.11b shows that all particles in the moving portion of the string move upward with constant *velocity* v_y , not constant acceleration. To see why this is so, we note that the *impulse* of the force F_y up to time t is $F_y t$. According to the impulse–momentum theorem (see Section 8.1), the impulse is equal to the change in the total transverse component of momentum of the moving part of the string. Because the system started with *zero* transverse momentum, this is equal to the total transverse momentum mv_y at time t :

$$\begin{aligned} \text{Transverse impulse} &= \text{Transverse momentum} \\ F_y t &= mv_y \end{aligned} \quad (15.13)$$

The total momentum thus must increase proportionately with time. But since the division point P moves with constant speed, the length of string that is in motion and hence the total mass m in motion are also proportional to the time t that the force has been acting. So the *change* of momentum must be associated entirely with the increasing amount of mass in motion, not with an increasing velocity of an individual mass element. That is, mv_y changes because m , not v_y , changes.

At time t , the left end of the string has moved up a distance $v_y t$, and the boundary point P has advanced a distance vt . The total force at the left end of the string has components F and F_y . Why F ? There is no motion in the direction along the length of the string, so there is no unbalanced horizontal force. Therefore F , the magnitude of the horizontal component, does not change when the string is displaced. In the displaced position the tension is $(F^2 + F_y^2)^{1/2}$; this is greater than F , so the string stretches somewhat.

To derive an expression for the wave speed v , we note that in Fig. 15.11b the right triangle whose vertex is at P , with sides $v_y t$ and vt , is similar to the right triangle whose vertex is at the position of the hand, with sides F_y and F . Hence

$$\frac{F_y}{F} = \frac{v_y t}{vt} \quad F_y = F \frac{v_y}{v}$$

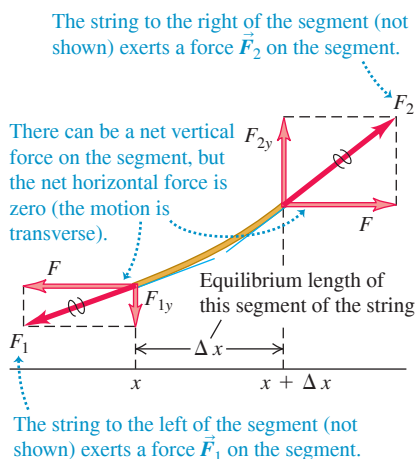
and

$$\text{Transverse impulse} = F_y t = F \frac{v_y}{v} t$$

Figure 15.12 These transmission cables have a relatively large amount of mass per unit length (μ) and a low tension (F). If the cables are disturbed—say, by a bird landing on them—transverse waves will travel along them at a slow speed $v = \sqrt{F/\mu}$.



Figure 15.13 Free-body diagram for a segment of string. The force at each end of the string is tangent to the string at the point of application.



The mass m of the moving portion of the string is the product of the mass per unit length μ and the length Δx , or $\mu \Delta x$. The transverse momentum is the product of this mass and the transverse velocity v_y :

$$\text{Transverse momentum} = mv_y = (\mu \Delta x)v_y$$

Substituting these into Eq. (15.13), we obtain

$$F \frac{v_y}{v} \Delta t = \mu \Delta x v_y$$

We solve this for the wave speed v :

$$\text{Speed of a transverse wave on a string} \quad v = \sqrt{\frac{F}{\mu}} \quad \begin{array}{l} \text{--- Tension in string} \\ \text{--- Mass per unit length} \end{array} \quad (15.14)$$

Equation (15.14) confirms that the wave speed v increases when the tension F increases but decreases when the mass per unit length μ increases (Fig. 15.12).

Note that v_y does not appear in Eq. (15.14); thus the wave speed doesn't depend on v_y . Our calculation considered only a very special kind of pulse, but we can consider *any* shape of wave disturbance as a series of pulses with different values of v_y . So even though we derived Eq. (15.14) for a special case, it is valid for *any* transverse wave motion on a string, including the sinusoidal and other periodic waves we discussed in Section 15.3. Note also that the wave speed doesn't depend on the amplitude or frequency of the wave, in accordance with our assumptions in Section 15.3.

Wave Speed on a String: Second Method

Here is an alternative derivation of Eq. (15.14). If you aren't comfortable with partial derivatives, it can be omitted. We apply Newton's second law, $\sum \vec{F} = m\vec{a}$, to a small segment of string whose length in the equilibrium position is Δx (Fig. 15.13). The mass of the segment is $m = \mu \Delta x$. The x -components of the forces have equal magnitude F and add to zero because the motion is transverse and there is no component of acceleration in the x -direction. To obtain F_{1y} and F_{2y} , we note that the ratio F_{1y}/F is equal in magnitude to the *slope* of the string at point x and that F_{2y}/F is equal to the slope at point $x + \Delta x$. Taking proper account of signs, we find

$$\frac{F_{1y}}{F} = -\left(\frac{\partial y}{\partial x}\right)_x \quad \frac{F_{2y}}{F} = \left(\frac{\partial y}{\partial x}\right)_{x+\Delta x} \quad (15.15)$$

The notation reminds us that the derivatives are evaluated at points x and $x + \Delta x$, respectively. From Eq. (15.15) we find that the net y -component of force is

$$F_y = F_{1y} + F_{2y} = F \left[\left(\frac{\partial y}{\partial x}\right)_{x+\Delta x} - \left(\frac{\partial y}{\partial x}\right)_x \right] \quad (15.16)$$

We now equate F_y from Eq. (15.16) to the mass $\mu \Delta x$ times the y -component of acceleration $\partial^2 y / \partial t^2$:

$$F \left[\left(\frac{\partial y}{\partial x}\right)_{x+\Delta x} - \left(\frac{\partial y}{\partial x}\right)_x \right] = \mu \Delta x \frac{\partial^2 y}{\partial t^2} \quad (15.17)$$

or, dividing Eq. (15.17) by $F \Delta x$,

$$\frac{\left(\frac{\partial y}{\partial x}\right)_{x+\Delta x} - \left(\frac{\partial y}{\partial x}\right)_x}{\Delta x} = \frac{\mu}{F} \frac{\partial^2 y}{\partial t^2} \quad (15.18)$$

We now take the limit as $\Delta x \rightarrow 0$. In this limit, the left side of Eq. (15.18) becomes the derivative of $\partial y / \partial x$ with respect to x (at constant t)—that is, the *second* (partial) derivative of y with respect to x :

$$\frac{\partial^2 y}{\partial x^2} = \frac{\mu}{F} \frac{\partial^2 y}{\partial t^2} \quad (15.19)$$

Now, Eq. (15.19) has exactly the same form as the *wave equation*, Eq. (15.12), that we derived at the end of Section 15.3. That equation and Eq. (15.19) describe the very same wave motion, so they must be identical. Comparing the two equations, we see that for this to be so, we must have

$$v = \sqrt{\frac{F}{\mu}}$$

which is the same expression as Eq. (15.14).

In going through this derivation, we didn't make any special assumptions about the shape of the wave. Since our derivation led us to rediscover Eq. (15.12), the wave equation, we conclude that the wave equation is valid for waves on a string that have *any* shape.

The Speed of Mechanical Waves

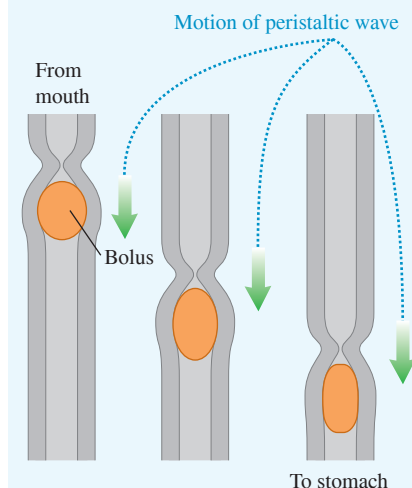
Equation (15.14) gives the wave speed for only the special case of mechanical waves on a stretched string or rope. Remarkably, it turns out that for many types of mechanical waves, including waves on a string, the expression for wave speed has the same general form:

$$v = \sqrt{\frac{\text{Restoring force returning the system to equilibrium}}{\text{Inertia resisting the return to equilibrium}}}$$

To interpret this expression, let's look at the now-familiar case of waves on a string. The tension F in the string plays the role of the restoring force; it tends to bring the string back to its undisturbed, equilibrium configuration. The mass of the string—or, more properly, the linear mass density μ —provides the inertia that prevents the string from returning instantaneously to equilibrium. Hence we have $v = \sqrt{F/\mu}$ for the speed of waves on a string.

In Chapter 16 we'll see a similar expression for the speed of sound waves in a gas. Roughly speaking, the gas pressure provides the force that tends to return the gas to its undisturbed state when a sound wave passes through. The inertia is provided by the density, or mass per unit volume, of the gas.

BIO APPLICATION Eating and Transverse Waves Swallowing food causes peristalsis, in which a transverse wave propagates down your esophagus. The wave is a radial contraction of the esophagus that pushes the bolus (the mass of swallowed food) toward the stomach. Unlike the speed of waves on a uniform string, the speed of this peristaltic wave is not constant: It averages about 3 cm/s in the upper esophagus, about 5 cm/s in the mid-esophagus, and about 2.5 cm/s in the lower esophagus.



EXAMPLE 15.3 Calculating wave speed

WITH VARIATION PROBLEMS

One end of a 2.00 kg rope is tied to a support at the top of a mine shaft 80.0 m deep (**Fig. 15.14**). The rope is stretched taut by a 20.0 kg box of rocks attached at the bottom. (a) A geologist at the bottom of the shaft signals to a colleague at the top by jerking the rope sideways. What is the speed of a transverse wave on the rope? (b) If a point on the rope is in transverse SHM with $f = 2.00$ Hz, how many cycles of the wave are there in the rope's length?

IDENTIFY and SET UP In part (a) we can find the wave speed (our target variable) by using the *dynamic* relationship $v = \sqrt{F/\mu}$ [Eq. (15.14)]. In part (b) we find the wavelength from the *kinematic* relationship $v = f\lambda$; from that we can find the target variable, the number of wavelengths that fit into the rope's 80.0 m length. We'll assume that the rope is massless (even though its weight is 10% that of the box), so that the box alone provides the tension in the rope.

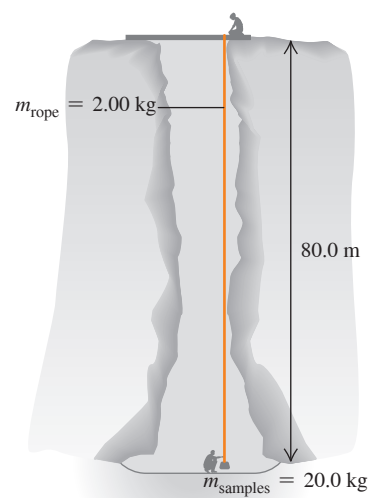
EXECUTE (a) The tension in the rope due to the box is

$$F = m_{\text{box}}g = (20.0 \text{ kg})(9.80 \text{ m/s}^2) = 196 \text{ N}$$

and the rope's linear mass density is

$$\mu = \frac{m_{\text{rope}}}{L} = \frac{2.00 \text{ kg}}{80.0 \text{ m}} = 0.0250 \text{ kg/m}$$

Figure 15.14 Using transverse waves to send signals along a vertical rope.



Continued

Hence, from Eq. (15.14), the wave speed is

$$v = \sqrt{\frac{F}{\mu}} = \sqrt{\frac{196 \text{ N}}{0.0250 \text{ kg/m}}} = 88.5 \text{ m/s}$$

(b) From Eq. (15.1), the wavelength is

$$\lambda = \frac{v}{f} = \frac{88.5 \text{ m/s}}{2.00 \text{ s}^{-1}} = 44.3 \text{ m}$$

There are $(80.0 \text{ m})/(44.3 \text{ m}) = 1.81$ wavelengths (that is, cycles of the wave) in the rope.

EVALUATE Because of the rope's weight, its tension is greater at the top than at the bottom. Hence both the wave speed and the wavelength increase as a wave travels up the rope. If you take account of this, can you verify that the wave speed at the top of the rope is 92.9 m/s?

KEYCONCEPT The speed of a transverse wave on a string is determined by the tension in the string and the string's linear mass density (mass per unit length). For a given string, the wave speed is independent of frequency or wavelength.

TEST YOUR UNDERSTANDING OF SECTION 15.4 The six strings of a guitar are the same length and under nearly the same tension, but they have different thicknesses. On which string do waves travel the fastest? (i) The thickest string; (ii) the thinnest string; (iii) the wave speed is the same on all strings.

ANSWER

(ii) The relationship $v = \sqrt{F/\mu}$ [Eq. (15.14)] says that the wave speed is greatest on the string with the smallest linear mass density. This is the thinnest string, which has the smallest amount of mass m and hence the smallest linear mass density $\mu = m/L$ (all strings are the same length).

BIO APPLICATION Surface Waves and the Swimming Speed of Ducks

When a duck swims, it necessarily produces waves on the surface of the water. The faster the duck swims, the larger the wave amplitude and the more power the duck must supply to produce these waves. The maximum power available from their leg muscles limits the maximum swimming speed of ducks to only about 0.7 m/s (2.5 km/h = 1.6 mi/h).

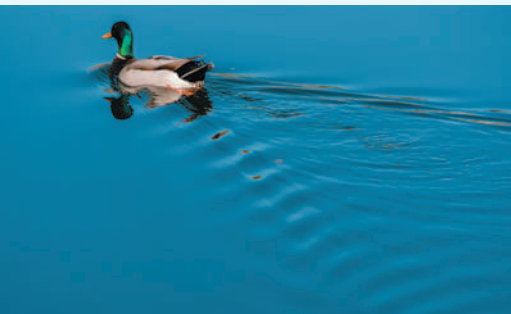
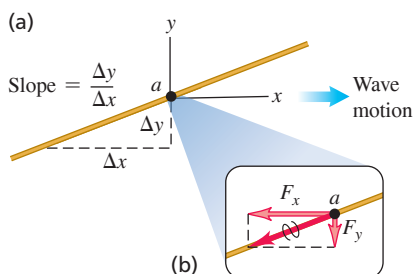


Figure 15.15 (a) Point a on a string carrying a wave from left to right. (b) The components of the force exerted on the part of the string to the right of point a by the part of the string to the left of point a .



15.5 ENERGY IN WAVE MOTION

Every wave motion has *energy* associated with it. The energy we receive from sunlight and the destructive effects of ocean surf and earthquakes bear this out. To produce any of the wave motions we have discussed in this chapter, we have to apply a force to a portion of the wave medium; the point where the force is applied moves, so we do *work* on the system. As the wave propagates, each portion of the medium exerts a force and does work on the adjoining portion. In this way a wave can transport energy from one region of space to another.

As an example, let's look again at transverse waves on a string. How is energy transferred from one portion of the string to another? Picture a wave traveling from left to right (the positive x -direction) past a point a on the string (Fig. 15.15a). The string to the left of point a exerts a force on the string to the right of it, and vice versa. In Fig. 15.15b we show the components F_x and F_y of the force that the string to the left of a exerts on the string to the right of a . As in Figs. 15.11 and 15.13, the magnitude of the horizontal component F_x equals the tension F in the undisturbed string. Note that F_y/F is equal to the negative of the *slope* of the string at a , and this slope is also given by $\partial y/\partial x$. Putting these together, we have

$$F_y(x, t) = -F \frac{\partial y(x, t)}{\partial x} \quad (15.20)$$

We need the negative sign because F_y is negative when the slope is positive (as in Fig. 15.15b). We write the vertical force as $F_y(x, t)$ as a reminder that its value may be different at different points along the string and at different times.

When point a moves in the y -direction, the force F_y does *work* on this point and therefore transfers energy into the part of the string to the right of a . The corresponding power P (rate of doing work) at the point a is the transverse force $F_y(x, t)$ at a times the transverse velocity $v_y(x, t) = \partial y(x, t)/\partial t$ of that point:

$$P(x, t) = F_y(x, t)v_y(x, t) = -F \frac{\partial y(x, t)}{\partial x} \frac{\partial y(x, t)}{\partial t} \quad (15.21)$$

This power is the *instantaneous* rate at which energy is transferred along the string at position x and time t . Note that energy is transferred only at points where the string has a nonzero slope ($\partial y/\partial x$ is nonzero), so that the tension force has a transverse component, and where the string has a nonzero transverse velocity ($\partial y/\partial t$ is nonzero) so that the transverse force can do work.

Equation (15.21) is valid for *any* wave on a string, sinusoidal or not. For a sinusoidal wave with wave function given by Eq. (15.7), we have

$$\begin{aligned} y(x, t) &= A \cos(kx - \omega t) \\ \frac{\partial y(x, t)}{\partial x} &= -kA \sin(kx - \omega t) \\ \frac{\partial y(x, t)}{\partial t} &= \omega A \sin(kx - \omega t) \\ P(x, t) &= Fk\omega A^2 \sin^2(kx - \omega t) \end{aligned} \quad (15.22)$$

By using the relationships $\omega = vk$ and $v^2 = F/\mu$, we can also express Eq. (15.22) in the alternative form

$$P(x, t) = \sqrt{\mu F} \omega^2 A^2 \sin^2(kx - \omega t) \quad (15.23)$$

The $\sin^2$ function is never negative, so the instantaneous power in a sinusoidal wave is either positive (so that energy flows in the positive x -direction) or zero (at points where there is no energy transfer). Energy is never transferred in the direction opposite to the direction of wave propagation (**Fig. 15.16**).

The maximum value of the instantaneous power $P(x, t)$ occurs when the $\sin^2$ function has the value unity:

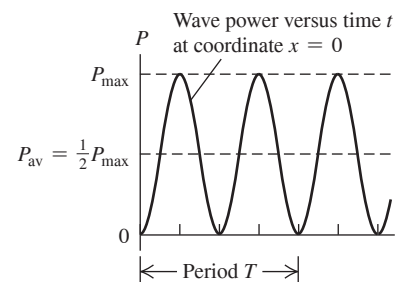
$$P_{\max} = \sqrt{\mu F} \omega^2 A^2 \quad (15.24)$$

The *average* value of the $\sin^2$ function, averaged over any whole number of cycles, is $\frac{1}{2}$. Hence we see from Eq. (15.23) that the *average* power P_{av} is just one-half the maximum instantaneous power $P_{\max}$ (Fig. 15.16):

$$P_{\text{av}} = \frac{1}{2} \sqrt{\mu F} \omega^2 A^2 \quad (15.25)$$

Average power, sinusoidal wave on a string Wave angular frequency Wave amplitude
Mass per unit length Tension in string

Figure 15.16 The instantaneous power $P(x, t)$ in a sinusoidal wave as given by Eq. (15.23), shown as a function of time at coordinate $x = 0$. The power is never negative, which means that energy never flows opposite to the direction of wave propagation.



The average rate of energy transfer is proportional to the square of the amplitude and to the square of the frequency. This proportionality is a general result for mechanical waves of all types, including seismic waves (see the photo that opens this chapter). For a mechanical wave, the rate of energy transfer quadruples if the frequency is doubled (for the same amplitude) or if the amplitude is doubled (for the same frequency).

Electromagnetic waves turn out to be a bit different. While the average rate of energy transfer in an electromagnetic wave is proportional to the square of the amplitude, just as for mechanical waves, it is independent of the value of ω .

EXAMPLE 15.4 Power in a wave

WITH VARIATION PROBLEMS

(a) In Example 15.2 (Section 15.3), at what maximum rate does Throcky put energy into the clothesline? That is, what is his maximum instantaneous power? The linear mass density of the clothesline is $\mu = 0.250$ kg/m, and Throcky applies tension $F = 36.0$ N. (b) What is his average power? (c) As Throcky tires, the amplitude decreases. What is the average power when the amplitude is 7.50 mm?

IDENTIFY and SET UP In part (a) our target variable is the *maximum instantaneous* power $P_{\max}$, while in parts (b) and (c) it is the *average* power. For part (a) we'll use Eq. (15.24), and for parts (b) and (c) we'll use Eq. (15.25); Example 15.2 gives us all the needed quantities.

EXECUTE (a) From Eq. (15.24),

$$\begin{aligned} P_{\max} &= \sqrt{\mu F} \omega^2 A^2 \\ &= \sqrt{(0.250 \text{ kg/m})(36.0 \text{ N})} (4.00\pi \text{ rad/s})^2 (0.075 \text{ m})^2 \\ &= 2.66 \text{ W} \end{aligned}$$

(b) From Eqs. (15.24) and (15.25), the average power is one-half of the maximum instantaneous power, so

$$P_{\text{av}} = \frac{1}{2} P_{\max} = \frac{1}{2} (2.66 \text{ W}) = 1.33 \text{ W}$$

Continued

(c) The new amplitude is $\frac{1}{10}$ of the value we used in parts (a) and (b). From Eq. (15.25), the average power is proportional to A^2 , so the new average power is

$$P_{\text{av}} = \left(\frac{1}{10}\right)^2(1.33 \text{ W}) = 0.0133 \text{ W} = 13.3 \text{ mW}$$

EVALUATE Equation (15.23) shows that P_{max} occurs when $\sin^2(kx - \omega t) = 1$. At any given position x , this happens twice per period of the wave—once when the sine function is equal to $+1$, and once when it's equal to -1 . The *minimum* instantaneous power is zero; this occurs when $\sin^2(kx - \omega t) = 0$, which also happens twice per period.

Can you confirm that the given values of μ and F give the wave speed mentioned in Example 15.2?

KEYCONCEPT The power (energy per second) carried by a wave on a string depends on the string tension and the linear mass density, as well as the frequency and amplitude of the wave. If the wave is sinusoidal, the average power equals one-half of the maximum instantaneous power.

Wave Intensity

Waves on a string carry energy in one dimension (along the direction of the string). But other types of waves, including sound waves in air and seismic waves within the earth, carry energy across all three dimensions of space. For waves of this kind, we define the **intensity** (denoted by I) to be *the time average rate at which energy is transported by the wave, per unit area*, across a surface perpendicular to the direction of propagation. Intensity I is average power per unit area and is usually measured in watts per square meter (W/m^2).

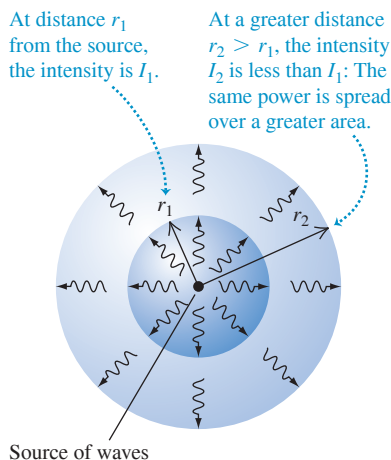
If waves spread out equally in all directions from a source, the intensity at a distance r from the source is inversely proportional to r^2 (**Fig. 15.17**). This result, called the *inverse-square law for intensity*, follows directly from energy conservation. If the power output of the source is P , then the average intensity I_1 through a sphere with radius r_1 and surface area $4\pi r_1^2$ is

$$I_1 = \frac{P}{4\pi r_1^2}$$

A similar expression gives the average intensity I_2 through a sphere with a different radius r_2 . If no energy is absorbed between the two spheres, the power P must be the same for both, and

$$4\pi r_1^2 I_1 = 4\pi r_2^2 I_2$$

Figure 15.17 The greater the distance from a wave source, the greater the area over which the wave power is distributed and the smaller the wave intensity.



Inverse-square law for intensity:

Intensity is inversely proportional to the square of the distance from source.

$$\begin{array}{l} \text{Intensity at point 1} \cdots I_1 \\ \text{Intensity at point 2} \cdots I_2 \end{array} = \frac{r_2^2}{r_1^2} \quad \begin{array}{l} \text{Distance from} \\ \text{source to point 2} \\ \text{Distance from} \\ \text{source to point 1} \end{array} \quad (15.26)$$

EXAMPLE 15.5 The inverse-square law

WITH VARIATION PROBLEMS

A siren on a tall pole radiates sound waves uniformly in all directions. At a distance of 15.0 m from the siren, the sound intensity is 0.250 W/m^2 . At what distance is the intensity 0.010 W/m^2 ?

IDENTIFY and SET UP Because sound is radiated uniformly in all directions, we can use the inverse-square law, Eq. (15.26). At $r_1 = 15.0 \text{ m}$ the intensity is $I_1 = 0.250 \text{ W/m}^2$, and the target variable is the distance r_2 at which the intensity is $I_2 = 0.010 \text{ W/m}^2$.

EXECUTE We solve Eq. (15.26) for r_2 :

$$r_2 = r_1 \sqrt{\frac{I_1}{I_2}} = (15.0 \text{ m}) \sqrt{\frac{0.250 \text{ W/m}^2}{0.010 \text{ W/m}^2}} = 75.0 \text{ m}$$

EVALUATE As a check on our answer, note that r_2 is five times greater than r_1 . By the inverse-square law, the intensity I_2 should be $1/5^2 = 1/25$ as great as I_1 , and indeed it is.

By using the inverse-square law, we've assumed that the sound waves travel in straight lines away from the siren. A more realistic solution, which is beyond our scope, would account for the reflection of sound waves from the ground.

KEYCONCEPT If waves emitted from a source spread out equally in all directions, the wave intensity is inversely proportional to the square of the distance from the source.

TEST YOUR UNDERSTANDING OF SECTION 15.5 Four identical strings each carry a sinusoidal wave of frequency 10 Hz. The string tension and wave amplitude are different for different strings. Rank the following strings in order from highest to lowest value of the average wave power: (i) tension 10 N, amplitude 1.0 mm; (ii) tension 40 N, amplitude 1.0 mm; (iii) tension 10 N, amplitude 4.0 mm; (iv) tension 20 N, amplitude 2.0 mm.

ANSWER

Equation (15.25) says that the average power in a sinusoidal wave on a string is $P_{av} = \frac{1}{2} \mu \omega^2 A^2 v$. The frequency f is the same for each wave, as is the angular frequency $\omega = 2\pi f$. Hence the average wave power for each string is proportional to the square root of the string tension F and the square of the amplitude A . Compared to string (i), the average power in each string is (ii) $\sqrt{4} = 2$ times greater; (iii) $4^2 = 16$ times greater; and (iv) $\sqrt{2}(2)^2 = 4\sqrt{2}$ times greater.

15.6 WAVE INTERFERENCE, BOUNDARY CONDITIONS, AND SUPERPOSITION

Up to this point we've been discussing waves that propagate continuously in the same direction. But when a wave strikes the boundaries of its medium, all or part of the wave is *reflected*. When you yell at a building wall or a cliff face some distance away, the sound wave is reflected from the rigid surface and you hear an echo. When you flip the end of a rope whose far end is tied to a rigid support, a pulse travels the length of the rope and is reflected back to you. In both cases, the initial and reflected waves overlap in the same region of the medium. We use the term **interference** to refer to what happens when two or more waves pass through the same region at the same time.

As a simple example of wave reflections and the role of the boundary of a wave medium, let's look again at transverse waves on a stretched string. What happens when a wave pulse or a sinusoidal wave arrives at the *end* of the string?

If the end is fastened to a rigid support as in **Fig. 15.18**, it is a *fixed end* that cannot move. The arriving wave exerts a force on the support (drawing 4 in Fig. 15.18); the reaction to this force, exerted by the support on the string, “kicks back” on the string and sets up a reflected pulse or wave traveling in the reverse direction (drawing 7). The reflected pulse moves in the opposite direction from the initial, or *incident*, pulse, and its displacement is also opposite.

The opposite situation from an end that is held stationary is a *free end*, one that is perfectly free to move in the direction perpendicular to the length of the string. For example, the string might be tied to a light ring that slides on a frictionless rod perpendicular to the string, as in **Fig. 15.19**. The ring and rod maintain the tension but exert no transverse force. When a wave arrives at this free end, the ring slides along the rod. The ring reaches a maximum displacement, and both it and the string come momentarily to rest, as in drawing 4 in Fig. 15.19. But the string is now stretched, giving increased tension, so the free end of the string is pulled back down, and again a reflected pulse is produced (drawing 7). As for a fixed end, the reflected pulse moves in the opposite direction from the initial pulse, but now the direction of the displacement is the same as for the initial pulse. The conditions at the end of the string, such as a rigid support or the complete absence of transverse force, are called **boundary conditions**.

The formation of the reflected pulse is similar to the overlap of two pulses traveling in opposite directions. **Figure 15.20** (next page) shows two pulses with the same shape, one inverted with respect to the other, traveling in opposite directions. As the pulses overlap and pass each other, the total displacement of the string is the *algebraic sum* of the displacements at that point in the individual pulses. Because these two pulses have the same shape, the total displacement at point *O* in the middle of the figure is zero at all times. Thus the motion of the left half of the string would be the same if we cut the string at point *O*, threw away the right side, and held the end at *O* fixed. The two pulses on the left side then correspond to the incident and reflected pulses, combining so that the total displacement at *O* is *always* zero. For this to occur, the reflected pulse must be inverted relative to the incident pulse, just as for reflection from the fixed end in Fig. 15.18.

Figure 15.18 Reflection of a wave pulse at a fixed end of a string. Time increases from top to bottom.

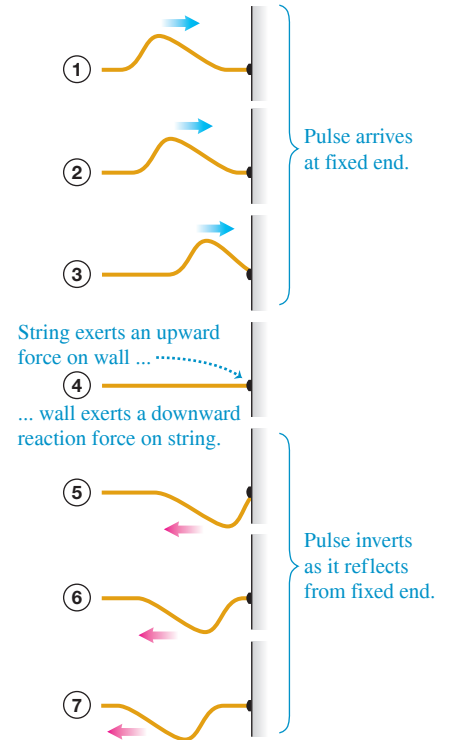


Figure 15.19 Reflection of a wave pulse at a free end of a string. Time increases from top to bottom. (Compare to Fig. 15.18.)

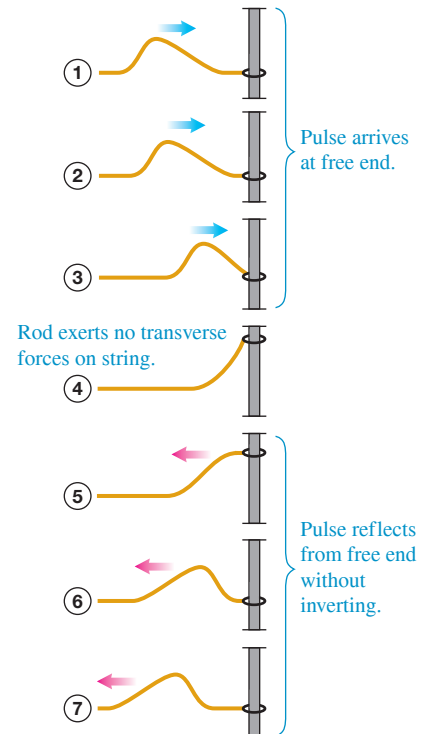


Figure 15.20 Overlap of two wave pulses—one right side up, one inverted—traveling in opposite directions. Time increases from top to bottom.

As the pulses overlap, the displacement of the string at any point is the algebraic sum of the displacements due to the individual pulses.

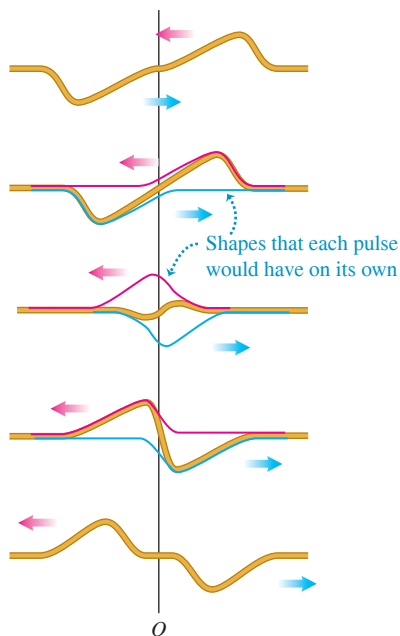


Figure 15.21 Overlap of two wave pulses—both right side up—traveling in opposite directions. Time increases from top to bottom. Compare to Fig. 15.20.

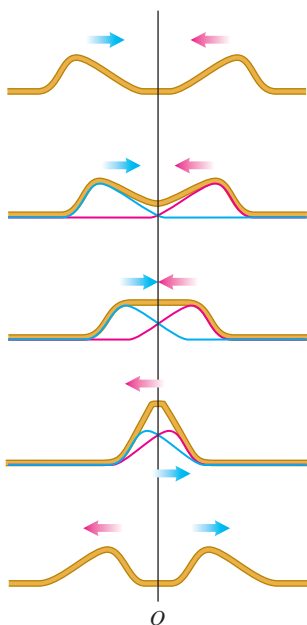


Figure 15.21 shows two pulses with the same shape, traveling in opposite directions but *not* inverted relative to each other. The displacement at point O in the middle of the figure is not zero, but the slope of the string at this point is always zero. According to Eq. (15.20), this corresponds to the absence of any transverse force at this point. In this case the motion of the left half of the string would be the same as if we cut the string at point O and attached the end to a frictionless sliding ring (Fig. 15.19) that maintains tension without exerting any transverse force. In other words, this situation corresponds to reflection of a pulse at a free end of a string at point O . In this case the reflected pulse is *not* inverted.

The Principle of Superposition

Combining the displacements of the separate pulses at each point to obtain the actual displacement is an example of the **principle of superposition**: When two waves overlap, the actual displacement of any point on the string at any time is obtained by adding the displacement the point would have if only the first wave were present and the displacement it would have if only the second wave were present. In other words, the wave function $y(x, t)$ for the resulting motion is obtained by *adding* the two wave functions for the two separate waves:

Principle of superposition: $y(x, t) = y_1(x, t) + y_2(x, t)$ (15.27)
 Wave function of combined wave = sum of individual wave functions

Mathematically, this additive property of wave functions follows from the form of the wave equation, Eq. (15.12) or (15.19), which every physically possible wave function must satisfy. Specifically, the wave equation is *linear*; that is, it contains the function $y(x, t)$ only to the first power (there are no terms involving $y(x, t)^2$, $y(x, t)^{1/2}$, etc.). As a result, if any two functions $y_1(x, t)$ and $y_2(x, t)$ satisfy the wave equation separately, their sum $y_1(x, t) + y_2(x, t)$ also satisfies it and is therefore a physically possible motion. Because this principle depends on the linearity of the wave equation and the corresponding linear-combination property of its solutions, it is also called the *principle of linear superposition*. For some physical systems, such as a medium that does not obey Hooke's law, the wave equation is *not* linear; this principle does not hold for such systems.

The principle of superposition is of central importance in all types of waves. When a friend talks to you while you are listening to music, you can distinguish the speech and the music from each other. This is precisely because the total sound wave reaching your ears is the algebraic sum of the wave produced by your friend's voice and the wave produced by the speakers of your stereo. If two sound waves did *not* combine in this simple linear way, the sound you would hear in this situation would be a hopeless jumble. Superposition also applies to electromagnetic waves (such as light).

TEST YOUR UNDERSTANDING OF SECTION 15.6

Figure 15.22 shows two wave pulses with different shapes traveling in different directions along a string. Make a series of sketches like Fig. 15.21 showing the shape of the string as the two pulses approach, overlap, and then pass each other.

ANSWER

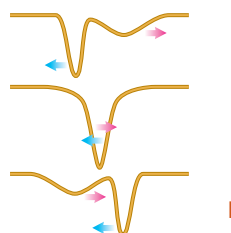
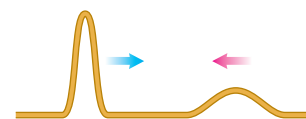


Figure 15.22 Two wave pulses with different shapes.



15.7 STANDING WAVES ON A STRING

We've looked at the reflection of a wave *pulse* on a string when it arrives at a boundary point (either a fixed end or a free end). Now let's consider what happens when a *sinusoidal* wave on a string is reflected by a fixed end. We'll again approach the problem by considering the superposition of two waves propagating through the string, one representing the incident wave and the other representing the wave reflected at the fixed end.

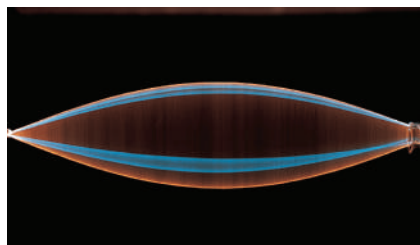
Figure 15.23 shows a string that is fixed at its left end. Its right end is moved up and down in simple harmonic motion to produce a wave that travels to the left; the wave reflected from the fixed end travels to the right. The resulting motion when the two waves combine no longer looks like two waves traveling in opposite directions. The string appears to be subdivided into segments, as in the time-exposure photographs of Figs. 15.23a, 15.23b, 15.23c, and 15.23d. Figure 15.23e shows two instantaneous shapes of the string in Fig. 15.23b. Let's compare this behavior with the waves we studied in Sections 15.1 through 15.5. In a wave that travels along the string, the amplitude is constant and the wave pattern moves with a speed equal to the wave speed. Here, instead, the wave pattern remains in the same position along the string and its amplitude fluctuates. There are particular points called **nodes** (labeled *N* in Fig. 15.23e) that never move at all. Midway between the nodes are points called **antinodes** (labeled *A* in Fig. 15.23e) where the amplitude of motion is greatest. Because the wave pattern doesn't appear to be moving in either direction along the string, it is called a **standing wave**. (To emphasize the difference, a wave that *does* move along the string is called a **traveling wave**.)

The principle of superposition explains how the incident and reflected waves combine to form a standing wave. In **Fig. 15.24** (next page) the red curves show a wave traveling to the left. The blue curves show a wave traveling to the right with the same propagation speed, wavelength, and amplitude. The waves are shown at nine instants, $\frac{1}{16}$ of a period apart. At each point along the string, we add the displacements (the values of *y*) for the two separate waves; the result is the total wave on the string, shown in gold.

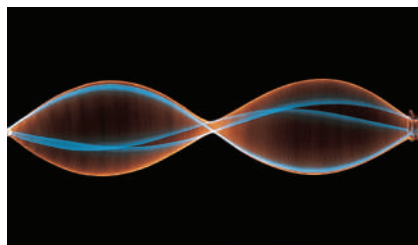
At certain instants, such as $t = \frac{1}{4}T$, the two wave patterns are exactly in phase with each other, and the shape of the string is a sine curve with twice the amplitude of either

Figure 15.23 (a)–(d) Time exposures of standing waves in a stretched string. From (a) to (d), the frequency of oscillation of the right-hand end increases and the wavelength of the standing wave decreases. (e) The extremes of the motion of the standing wave in part (b), with nodes at the center and at the ends. The right-hand end of the string moves very little compared to the antinodes and so is essentially a node.

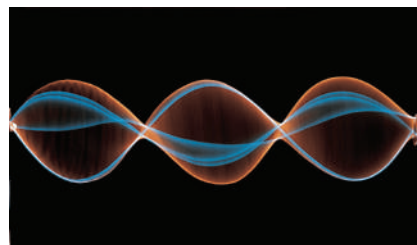
(a) String is one-half wavelength long.



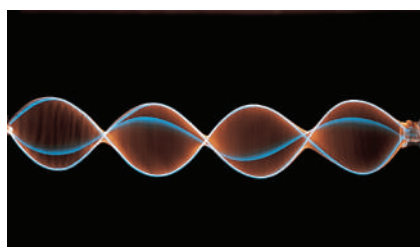
(b) String is one wavelength long.



(c) String is one and a half wavelengths long.



(d) String is two wavelengths long.



(e) The shape of the string in (b) at two different instants

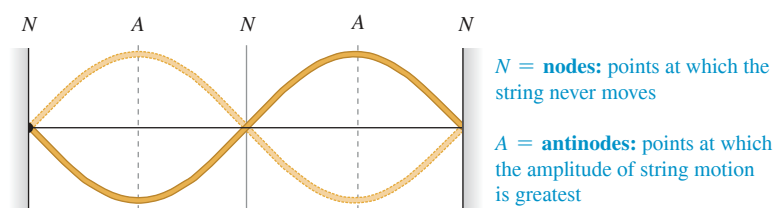
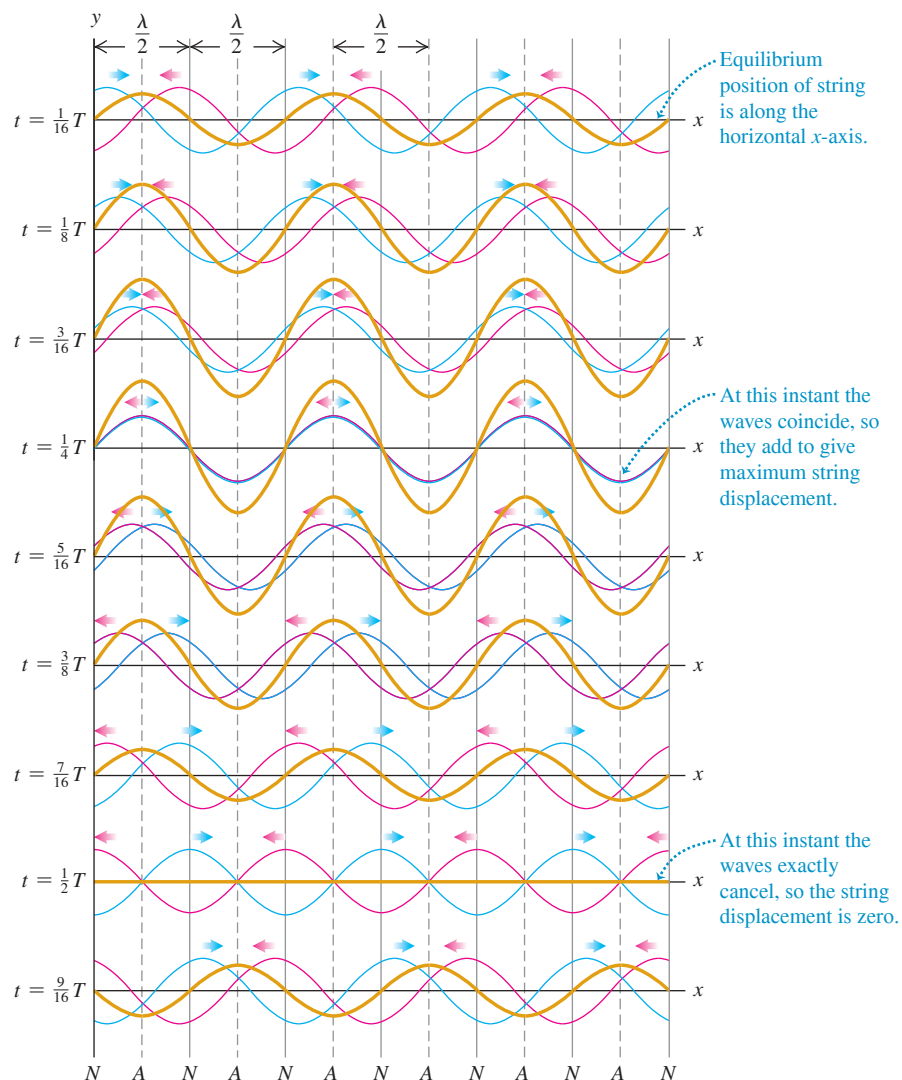


Figure 15.24 Formation of a standing wave. A wave traveling to the left (red curves) combines with a wave traveling to the right (blue curves) to form a standing wave (gold curves).



individual wave. At other instants, such as $t = \frac{1}{2}T$, the two waves are exactly out of phase with each other, and the total wave at that instant is zero. The resultant displacement is *always* zero at those places marked N at the bottom of Fig. 15.24. These are the **nodes**. At a node the displacements of the two waves in red and blue are always equal and opposite and cancel each other out. This cancellation is called **destructive interference**. Midway between the nodes are the points of *greatest* amplitude, or the **antinodes**, marked A . At the antinodes the displacements of the two waves in red and blue are always identical, giving a large resultant displacement; this phenomenon is called **constructive interference**. We can see from the figure that the distance between successive nodes or between successive antinodes is one half-wavelength, or $\lambda/2$.

We can derive a wave function for the standing wave of Fig. 15.24 by adding the wave functions $y_1(x, t)$ and $y_2(x, t)$ for two waves with equal amplitude, period, and wavelength traveling in opposite directions. Here $y_1(x, t)$ (the red curves in Fig. 15.24) represents an incoming, or *incident*, wave traveling to the left along the $+x$ -axis, arriving at the point $x = 0$ and being reflected; $y_2(x, t)$ (the blue curves in Fig. 15.24) represents the *reflected* wave traveling to the right from $x = 0$. We noted in Section 15.6 that the wave reflected from a fixed end of a string is inverted, so we give a negative sign to one of the waves:

$$y_1(x, t) = -A \cos(kx + \omega t) \quad (\text{incident wave traveling to the left})$$

$$y_2(x, t) = A \cos(kx - \omega t) \quad (\text{reflected wave traveling to the right})$$

The change in sign corresponds to a shift in *phase* of 180° or π radians. At $x = 0$ the motion from the reflected wave is $A \cos \omega t$ and the motion from the incident wave is $-A \cos \omega t$, which we can also write as $A \cos(\omega t + \pi)$. From Eq. (15.27), the wave function for the standing wave is the sum of the individual wave functions:

$$y(x, t) = y_1(x, t) + y_2(x, t) = A[-\cos(kx + \omega t) + \cos(kx - \omega t)]$$

We can rewrite each of the cosine terms by using the identities for the cosine of the sum and difference of two angles: $\cos(a \pm b) = \cos a \cos b \mp \sin a \sin b$. Applying these and combining terms, we obtain the wave function for the standing wave:

$$y(x, t) = y_1(x, t) + y_2(x, t) = (2A \sin kx) \sin \omega t \quad \text{or}$$

Standing wave on a string, fixed end at $x = 0$:

$$y(x, t) = (A_{\text{SW}} \sin kx) \sin \omega t \quad (15.28)$$

Wave function
Standing-wave amplitude
Time
Wave number
Position
Angular frequency

The standing-wave amplitude A_{SW} is twice the amplitude A of either of the original traveling waves: $A_{\text{SW}} = 2A$.

Equation (15.28) has two factors: a function of x and a function of t . The factor $A_{\text{SW}} \sin kx$ shows that at each instant the shape of the string is a sine curve. But unlike a wave traveling along a string, the wave shape stays in the same position, oscillating up and down as described by the $\sin \omega t$ factor. This behavior is shown by the gold curves in Fig. 15.24. Each point in the string still undergoes simple harmonic motion, but all the points between any successive pair of nodes oscillate *in phase*. This is in contrast to the phase differences between oscillations of adjacent points that we see with a traveling wave.

We can use Eq. (15.28) to find the positions of the nodes; these are the points for which $\sin kx = 0$, so the displacement is *always* zero. This occurs when $kx = 0, \pi, 2\pi, 3\pi, \dots$, or, using $k = 2\pi/\lambda$,

$$\begin{aligned} x &= 0, \frac{\pi}{k}, \frac{2\pi}{k}, \frac{3\pi}{k}, \dots \\ &= 0, \frac{\lambda}{2}, \frac{2\lambda}{2}, \frac{3\lambda}{2}, \dots \end{aligned} \quad \begin{array}{l} \text{(nodes of a standing wave on} \\ \text{a string, fixed end at } x = 0) \end{array} \quad (15.29)$$

In particular, there is a node at $x = 0$, as there should be, since this point is a fixed end of the string.

A standing wave, unlike a traveling wave, *does not* transfer energy from one end to the other. The two waves that form it would individually carry equal amounts of power in opposite directions. There is a local flow of energy from each node to the adjacent antinodes and back, but the *average* rate of energy transfer is zero at every point. If you use the wave function of Eq. (15.28) to evaluate the wave power given by Eq. (15.21), you will find that the average power is zero.

PROBLEM-SOLVING STRATEGY 15.2 Standing Waves

IDENTIFY *the relevant concepts:* Identify the target variables. Then determine whether the problem is purely *kinematic* (involving only such quantities as wave speed v , wavelength λ , and frequency f) or whether *dynamic* properties of the medium (such as F and μ for transverse waves on a string) are also involved.

SET UP *the problem* using the following steps:

1. Sketch the shape of the standing wave at a particular instant. This will help you visualize the nodes (label them N) and antinodes (A). The distance between adjacent nodes (or antinodes) is $\lambda/2$; the distance between a node and the adjacent antinode is $\lambda/4$.
2. Choose the equations you'll use. The wave function for the standing wave, like Eq. (15.28), is often useful.

3. You can determine the wave speed if you know λ and f (or, equivalently, $k = 2\pi/\lambda$ and $\omega = 2\pi f$) or if you know the relevant properties of the medium (for a string, F and μ).

EXECUTE *the solution:* Solve for the target variables. Once you've found the wave function, you can find the displacement y at any point x and at any time t . You can find the velocity and acceleration of a particle in the medium by taking the first and second partial derivatives of y with respect to time.

EVALUATE *your answer:* Compare your numerical answers with your sketch. Check that the wave function satisfies the boundary conditions (for example, the displacement should be zero at a fixed end).

EXAMPLE 15.6 Standing waves on a guitar string**WITH VARIATION PROBLEMS**

A guitar string lies along the x -axis when in equilibrium. The end of the string at $x = 0$ (the bridge of the guitar) is fixed. A sinusoidal wave with amplitude $A = 0.750 \text{ mm} = 7.50 \times 10^{-4} \text{ m}$ and frequency $f = 440 \text{ Hz}$, corresponding to the red curves in Fig. 15.24, travels along the string in the $-x$ -direction at 143 m/s . It is reflected from the fixed end, and the superposition of the incident and reflected waves forms a standing wave. (a) Find the equation giving the displacement of a point on the string as a function of position and time. (b) Locate the nodes. (c) Find the amplitude of the standing wave and the maximum transverse velocity and acceleration.

IDENTIFY and SET UP This is a *kinematics* problem (see Problem-Solving Strategy 15.1 in Section 15.3). The target variables are: in part (a), the wave function of the standing wave; in part (b), the locations of the nodes; and in part (c), the maximum displacement y , transverse velocity v_y , and transverse acceleration a_y . Since there is a fixed end at $x = 0$, we can use Eqs. (15.28) and (15.29) to describe this standing wave. We'll need the relationships $\omega = 2\pi f$, $v = \omega/k$, and $v = \lambda f$.

EXECUTE (a) The standing-wave amplitude is $A_{\text{SW}} = 2A = 1.50 \times 10^{-3} \text{ m}$ (twice the amplitude of either the incident or reflected wave). The angular frequency and wave number are

$$\omega = 2\pi f = (2\pi \text{ rad})(440 \text{ s}^{-1}) = 2760 \text{ rad/s}$$

$$k = \frac{\omega}{v} = \frac{2760 \text{ rad/s}}{143 \text{ m/s}} = 19.3 \text{ rad/m}$$

Equation (15.28) then gives

$$\begin{aligned} y(x, t) &= (A_{\text{SW}} \sin kx) \sin \omega t \\ &= [(1.50 \times 10^{-3} \text{ m}) \sin(19.3 \text{ rad/m})x] \sin(2760 \text{ rad/s})t \end{aligned}$$

(b) From Eq. (15.29), the positions of the nodes are $x = 0, \lambda/2, \lambda, 3\lambda/2, \dots$. The wavelength is $\lambda = v/f = (143 \text{ m/s})/(440 \text{ Hz}) = 0.325 \text{ m}$, so the nodes are at $x = 0, 0.163 \text{ m}, 0.325 \text{ m}, 0.488 \text{ m}, \dots$

(c) From the expression for $y(x, t)$ in part (a), the maximum displacement from equilibrium is $A_{\text{SW}} = 1.50 \times 10^{-3} \text{ m} = 1.50 \text{ mm}$.

This occurs at the *antinodes*, which are midway between adjacent nodes (that is, at $x = 0.081 \text{ m}, 0.244 \text{ m}, 0.406 \text{ m}, \dots$).

For a particle on the string at any point x , the transverse (y -) velocity is

$$\begin{aligned} v_y(x, t) &= \frac{\partial y(x, t)}{\partial t} \\ &= [(1.50 \times 10^{-3} \text{ m}) \sin(19.3 \text{ rad/m})x] \\ &\quad \times [(2760 \text{ rad/s}) \cos(2760 \text{ rad/s})t] \\ &= [(4.15 \text{ m/s}) \sin(19.3 \text{ rad/m})x] \cos(2760 \text{ rad/s})t \end{aligned}$$

At an antinode, $\sin(19.3 \text{ rad/m})x = \pm 1$ and the transverse velocity varies between $+4.15 \text{ m/s}$ and -4.15 m/s . As is always the case in SHM, the maximum velocity occurs when the particle is passing through the equilibrium position ($y = 0$).

The transverse acceleration $a_y(x, t)$ is the *second* partial derivative of $y(x, t)$ with respect to time. You can show that

$$\begin{aligned} a_y(x, t) &= \frac{\partial v_y(x, t)}{\partial t} = \frac{\partial^2 y(x, t)}{\partial t^2} \\ &= [(-1.15 \times 10^4 \text{ m/s}^2) \sin(19.3 \text{ rad/m})x] \sin(2760 \text{ rad/s})t \end{aligned}$$

At the antinodes, the transverse acceleration varies between $+1.15 \times 10^4 \text{ m/s}^2$ and $-1.15 \times 10^4 \text{ m/s}^2$.

EVALUATE The maximum transverse velocity at an antinode is quite respectable (about 15 km/h , or 9.3 mi/h). But the maximum transverse acceleration is tremendous, 1170 times the acceleration due to gravity! Guitar strings are actually fixed at *both* ends; we'll see the consequences of this in the next section.

KEYCONCEPT A sinusoidal standing wave is the superposition of two sinusoidal waves of the same amplitude and frequency traveling in opposite directions. The wave nodes, which are spaced one half-wavelength apart, are points where the displacement in the standing wave is always zero.

TEST YOUR UNDERSTANDING OF SECTION 15.7 Suppose the frequency of the standing wave in Example 15.6 were doubled from 440 Hz to 880 Hz . Would all of the nodes for $f = 440 \text{ Hz}$ also be nodes for $f = 880 \text{ Hz}$? If so, would there be additional nodes for $f = 880 \text{ Hz}$? If not, which nodes are absent for $f = 880 \text{ Hz}$?

ANSWER

Yes, yes Doubling the frequency makes the wavelength half as large. Hence the spacing between nodes (equal to $\lambda/2$) is also half as large. There are nodes at all of the previous positions, but there is also a new node between every pair of old nodes.

15.8 NORMAL MODES OF A STRING

When we described standing waves on a string rigidly held at one end, as in Fig. 15.23, we made no assumptions about the length of the string or about what was happening at the other end. Let's now consider a string of a definite length L , rigidly held at *both* ends. Such strings are found in many musical instruments, including pianos, violins, and guitars. When a guitar string is plucked, a wave is produced in the string; this wave is reflected and re-reflected from the ends of the string, making a standing wave. This standing wave on the string in turn produces a sound wave in the air, with a frequency determined by the properties of the string. This is what makes stringed instruments so useful in making music.

To understand a standing wave on a string fixed at both ends, we first note that the standing wave must have a node at *both* ends of the string. We saw in the preceding section that adjacent nodes are one half-wavelength ($\lambda/2$) apart, so the length of the string must be $\lambda/2$, or $2(\lambda/2)$, or $3(\lambda/2)$, or in general some integer number of half-wavelengths:

$$L = n \frac{\lambda}{2} \quad (n = 1, 2, 3, \dots) \quad (\text{string fixed at both ends}) \quad (15.30)$$

That is, if a string with length L is fixed at both ends, a standing wave can exist only if its wavelength satisfies Eq. (15.30).

Solving this equation for λ and labeling the possible values of λ as λ_n , we find

$$\lambda_n = \frac{2L}{n} \quad (n = 1, 2, 3, \dots) \quad (\text{string fixed at both ends}) \quad (15.31)$$

Waves can exist on the string if the wavelength is *not* equal to one of these values, but there cannot be a steady wave pattern with nodes and antinodes, and the total wave cannot be a standing wave. Equation (15.31) is illustrated by the standing waves shown in Figs. 15.23a, 15.23b, 15.23c, and 15.23d; these represent $n = 1, 2, 3$, and 4, respectively.

Corresponding to the series of possible standing-wave wavelengths λ_n is a series of possible standing-wave frequencies f_n , each related to its corresponding wavelength by $f_n = v/\lambda_n$. The smallest frequency f_1 corresponds to the largest wavelength (the $n = 1$ case), $\lambda_1 = 2L$:

$$f_1 = \frac{v}{2L} \quad (\text{string fixed at both ends}) \quad (15.32)$$

This is called the **fundamental frequency**. The other standing-wave frequencies are $f_2 = 2v/2L$, $f_3 = 3v/2L$, and so on. These are all integer multiples of f_1 , such as $2f_1$, $3f_1$, $4f_1$, and so on. We can express *all* the frequencies as

$$\text{Standing-wave frequencies, string fixed at both ends: } f_n = n \frac{v}{2L} = n f_1 \quad (n = 1, 2, 3, \dots) \quad (15.33)$$

Wave speed Fundamental frequency = $v/2L$
Length of string

These frequencies are called **harmonics**, and the series is called a **harmonic series**. Musicians sometimes call f_2 , f_3 , and so on **overtones**; f_2 is the second harmonic or the first overtone, f_3 is the third harmonic or the second overtone, and so on. The first harmonic is the same as the fundamental frequency (**Fig. 15.25**).

For a string with fixed ends at $x = 0$ and $x = L$, the wave function $y(x, t)$ of the n th standing wave is given by Eq. (15.28) (which satisfies the condition that there is a node at $x = 0$), with $\omega = \omega_n = 2\pi f_n$ and $k = k_n = 2\pi/\lambda_n$:

$$y_n(x, t) = A_{\text{SW}} \sin k_n x \sin \omega_n t \quad (15.34)$$

You can confirm that this wave function has nodes at both $x = 0$ and $x = L$.

A **normal mode** of an oscillating system is a motion in which all particles of the system move sinusoidally with the same frequency. For a system made up of a string of length L fixed at both ends, each of the frequencies given by Eq. (15.33) corresponds to a possible normal-mode pattern. **Figure 15.26** shows the first four normal-mode patterns and their associated frequencies and wavelengths; these correspond to Eq. (15.34) with $n = 1, 2, 3$, and 4. By contrast, a harmonic oscillator, which has only one oscillating particle, has only one normal mode and one characteristic frequency. The string fixed at both ends has infinitely many normal modes ($n = 1, 2, 3, \dots$) because it is made up of a very large (effectively infinite) number of particles. More complicated oscillating systems also have infinite numbers of normal modes, though with more complex normal-mode patterns (**Fig. 15.27**, next page).

Figure 15.25 Each string of a violin naturally oscillates at its harmonic frequencies, producing sound waves in the air with the same frequencies.



Figure 15.26 The first four normal modes of a string fixed at both ends. (Compare these to the photographs in Fig. 15.23.)

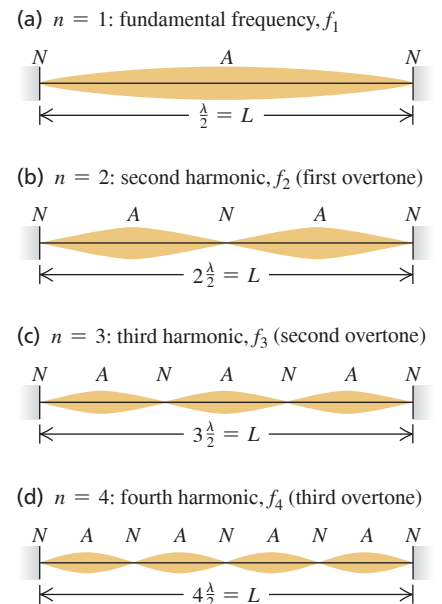


Figure 15.27 Astronomers have discovered that the sun oscillates in several different normal modes. This computer simulation shows one such mode.

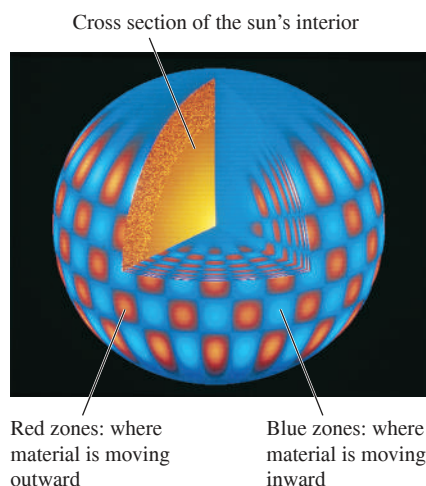
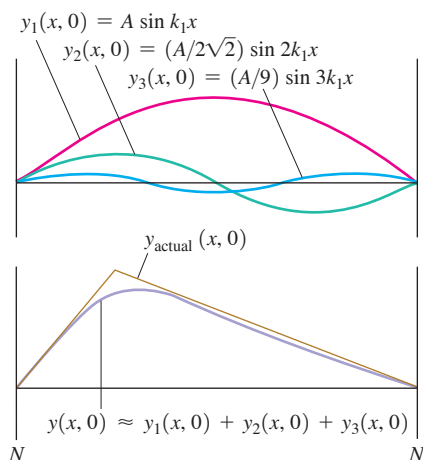


Figure 15.28 When a guitar string is plucked (pulled into a triangular shape) and released, a standing wave results. The standing wave is well represented (except at the sharp maximum point) by the sum of just three sinusoidal functions. Including additional sinusoidal functions further improves the representation.



Complex Standing Waves

If we could displace a string so that its shape is the same as one of the normal-mode patterns and then release it, it would vibrate with the frequency of that mode. Such a vibrating string would displace the surrounding air with the same frequency, producing a traveling sinusoidal sound wave that your ears would perceive as a pure tone. But when a string is struck (as in a piano) or plucked (as is done to guitar strings), the shape of the displaced string is *not* one of the patterns in Fig. 15.26. The motion is therefore a combination or *superposition* of many normal modes. Several simple harmonic motions of different frequencies are present simultaneously, and the displacement of any point on the string is the superposition of the displacements associated with the individual modes. The sound produced by the vibrating string is likewise a superposition of traveling sinusoidal sound waves, which you perceive as a rich, complex tone with the fundamental frequency f_1 . The standing wave on the string and the traveling sound wave in the air have similar **harmonic content** (the extent to which frequencies higher than the fundamental are present). The harmonic content depends on how the string is initially set into motion. If you pluck the strings of an acoustic guitar in the normal location over the sound hole, the sound that you hear has a different harmonic content than if you pluck the strings next to the fixed end on the guitar body.

It is possible to represent every possible motion of the string as some superposition of normal-mode motions. Finding this representation for a given vibration pattern is called *harmonic analysis*. The sum of sinusoidal functions that represents a complex wave is called a *Fourier series*. **Figure 15.28** shows how a standing wave that is produced by plucking a guitar string of length L at a point $L/4$ from one end can be represented as a combination of sinusoidal functions.

Standing Waves and String Instruments

From Eq. (15.32), the fundamental frequency of a vibrating string is $f_1 = v/2L$. The speed v of waves on the string is determined by Eq. (15.14), $v = \sqrt{F/\mu}$. Combining these equations, we find

$$\text{Fundamental frequency, string fixed at both ends} \rightarrow f_1 = \frac{1}{2L} \sqrt{\frac{F}{\mu}} \quad (15.35)$$

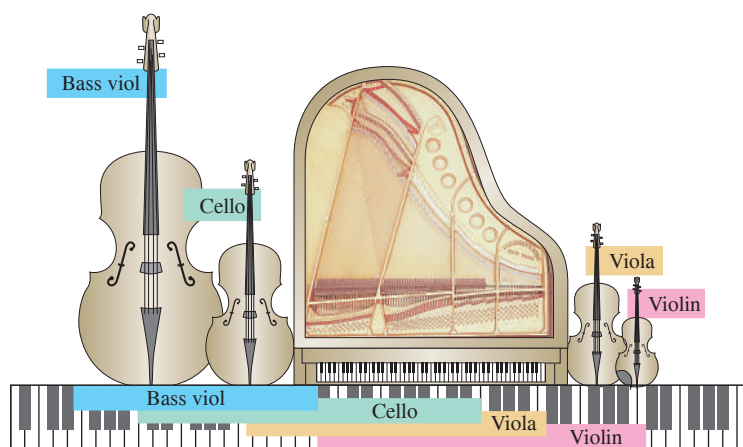
← Tension in string
← Mass per unit length
← Length of string

This is also the fundamental frequency of the sound wave created in the surrounding air by the vibrating string. The inverse dependence of frequency on length L is

illustrated by the long strings of the bass (low-frequency) section of the piano or the bass viol compared with the shorter strings of the treble section of the piano or the violin (**Fig. 15.29**). The pitch of a violin or guitar is usually varied by pressing a string against the fingerboard with the fingers to change the length L of the vibrating portion of the string. Increasing the tension F increases the wave speed v and thus increases the frequency (and the pitch). All string instruments are “tuned” to the correct frequencies by varying the tension; you tighten the string to raise the pitch. Finally, increasing the mass per unit length μ decreases the wave speed and thus the frequency. The lower notes on a steel guitar are produced by thicker strings, and one reason for winding the bass strings of a piano with wire is to obtain the desired low frequency from a relatively short string.

Wind instruments such as saxophones and trombones also have normal modes. As for stringed instruments, the frequencies of these normal modes determine the pitch of the musical tones that these instruments produce. We’ll discuss these instruments and many other aspects of sound in Chapter 16.

Figure 15.29 Comparing the range of a concert grand piano to the ranges of a bass viol, a cello, a viola, and a violin. In all cases, longer strings produce bass notes and shorter strings produce treble notes.



EXAMPLE 15.7 A giant bass viol

WITH VARIATION PROBLEMS

In an attempt to get your name in *Guinness World Records*, you build a bass viol with strings of length 5.00 m between fixed points. One string, with linear mass density 40.0 g/m, is tuned to a 20.0 Hz fundamental frequency (the lowest frequency that the human ear can hear). Calculate (a) the tension of this string, (b) the frequency and wavelength on the string of the second harmonic, and (c) the frequency and wavelength on the string of the second overtone.

IDENTIFY and SET UP In part (a) the target variable is the string tension F ; we’ll use Eq. (15.35), which relates F to the known values $f_1 = 20.0$ Hz, $L = 5.00$ m, and $\mu = 40.0$ g/m. In parts (b) and (c) the target variables are the frequency and wavelength of a given harmonic and a given overtone. We determine these from the given length of the string and the fundamental frequency, using Eqs. (15.31) and (15.33).

EXECUTE (a) We solve Eq. (15.35) for F :

$$F = 4\mu L^2 f_1^2 = 4(40.0 \times 10^{-3} \text{ kg/m})(5.00 \text{ m})^2(20.0 \text{ s}^{-1})^2 \\ = 1600 \text{ N} = 360 \text{ lb}$$

(b) From Eqs. (15.33) and (15.31), the frequency and wavelength of the second harmonic ($n = 2$) are

$$f_2 = 2f_1 = 2(20.0 \text{ Hz}) = 40.0 \text{ Hz}$$

$$\lambda_2 = \frac{2L}{2} = \frac{2(5.00 \text{ m})}{2} = 5.00 \text{ m}$$

(c) The second overtone is the “second tone over” (above) the fundamental—that is, $n = 3$. Its frequency and wavelength are

$$f_3 = 3f_1 = 3(20.0 \text{ Hz}) = 60.0 \text{ Hz}$$

$$\lambda_3 = \frac{2L}{3} = \frac{2(5.00 \text{ m})}{3} = 3.33 \text{ m}$$

EVALUATE The string tension in a real bass viol is typically a few hundred newtons; the tension in part (a) is a bit higher than that. The wavelengths in parts (b) and (c) are equal to the length of the string and two-thirds the length of the string, respectively, which agrees with the drawings of standing waves in Fig. 15.26.

KEYCONCEPT A string with its ends fixed vibrates in a type of standing wave called a normal mode when all of its particles move sinusoidally with the same frequency. The frequencies of oscillation of the normal modes are integer multiples of a minimum normal-mode frequency, called the fundamental frequency.

EXAMPLE 15.8 From waves on a string to sound waves in air**WITH VARIATION PROBLEMS**

What are the frequency and wavelength of the sound waves produced in the air when the string in Example 15.7 is vibrating at its fundamental frequency? The speed of sound in air at 20°C is 344 m/s.

IDENTIFY and SET UP Our target variables are the frequency and wavelength for the *sound wave* produced by the bass viol string. The frequency of the sound wave is the same as the fundamental frequency f_1 of the standing wave, because the string forces the surrounding air to vibrate at the same frequency. The wavelength of the sound wave is $\lambda_{1(\text{sound})} = v_{\text{sound}}/f_1$.

EXECUTE We have $f = f_1 = 20.0$ Hz, so

$$\lambda_{1(\text{sound})} = \frac{v_{\text{sound}}}{f_1} = \frac{344 \text{ m/s}}{20.0 \text{ Hz}} = 17.2 \text{ m}$$

EVALUATE In Example 15.7, the wavelength of the fundamental on the string was $\lambda_{1(\text{string})} = 2L = 2(5.00 \text{ m}) = 10.0 \text{ m}$. Here $\lambda_{1(\text{sound})} = 17.2 \text{ m}$ is greater than that by the factor of $17.2/10.0 = 1.72$. This is as it should be: Because the frequencies of the sound wave and the standing wave are equal, $\lambda = v/f$ says that the wavelengths in air and on the string are in the same ratio as the corresponding wave speeds; here $v_{\text{sound}} = 344 \text{ m/s}$ is greater than $v_{\text{string}} = (10.0 \text{ m})(20.0 \text{ Hz}) = 200 \text{ m/s}$ by just the factor 1.72.

KEYCONCEPT A string vibrating at a certain frequency produces sound waves of the same frequency in the surrounding air. The standing wave on the string and the sound wave will have different wavelengths, however, if the speed of waves on the string does not equal the speed of sound.

TEST YOUR UNDERSTANDING OF SECTION 15.8 While a guitar string is vibrating, you gently touch the midpoint of the string to ensure that the string does not vibrate at that point. Which normal modes *cannot* be present on the string while you are touching it in this way?

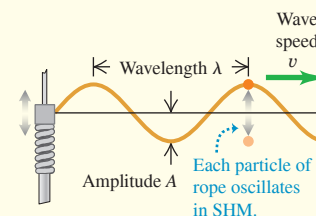
ANSWER $n = 1, 3, 5, \dots$. When you touch the string at its center, you are producing a node at the center. Hence only standing waves with a node at $x = L/2$ are allowed. From Figure 15.26 you can see that the normal modes $n = 1, 3, 5, \dots$ cannot be present.

CHAPTER 15 SUMMARY

Waves and their properties: A wave is any disturbance that propagates from one region to another. A mechanical wave travels within some material called the medium. The wave speed v depends on the type of wave and the properties of the medium.

In a periodic wave, the motion of each point of the medium is periodic with frequency f and period T . The wavelength λ is the distance over which the wave pattern repeats, and the amplitude A is the maximum displacement of a particle in the medium. The product of λ and f equals the wave speed. A sinusoidal wave is a special periodic wave in which each point moves in simple harmonic motion. (See Example 15.1.)

$$v = \lambda f \quad (15.1)$$



Wave functions and wave dynamics: The wave function $y(x, t)$ describes the displacements of individual particles in the medium. Equations (15.3), (15.4), and (15.7) give the wave equation for a sinusoidal wave traveling in the $+x$ -direction. If the wave is moving in the $-x$ -direction, the minus signs in the cosine functions are replaced by plus signs. (See Example 15.2.)

The wave function obeys a partial differential equation called the wave equation, Eq. (15.12).

The speed of transverse waves on a string depends on the tension F and mass per unit length μ . (See Example 15.3.)

$$y(x, t) = A \cos \left[\omega \left(\frac{x}{v} - t \right) \right] \quad (15.3)$$

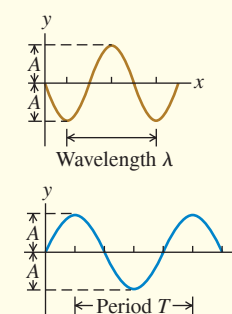
$$y(x, t) = A \cos 2\pi \left[\left(\frac{x}{\lambda} - \frac{t}{T} \right) \right] \quad (15.4)$$

$$y(x, t) = A \cos(kx - \omega t) \quad (15.7)$$

where $k = 2\pi/\lambda$ and $\omega = 2\pi f = vk$

$$\frac{\partial^2 y(x, t)}{\partial x^2} = \frac{1}{v^2} \frac{\partial^2 y(x, t)}{\partial t^2} \quad (15.12)$$

$$v = \sqrt{\frac{F}{\mu}} \text{ (waves on a string)} \quad (15.14)$$



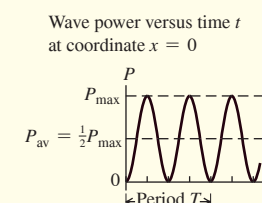
Wave power: Wave motion conveys energy from one region to another. For a sinusoidal mechanical wave, the average power P_{av} is proportional to the square of the wave amplitude and the square of the frequency. For waves that spread out in three dimensions, the wave intensity I is inversely proportional to the square of the distance from the source. (See Examples 15.4 and 15.5.)

$$P_{av} = \frac{1}{2} \sqrt{\mu F} \omega^2 A^2 \quad (15.25)$$

(average power, sinusoidal wave)

$$\frac{I_1}{I_2} = \frac{r_2^2}{r_1^2} \quad (15.26)$$

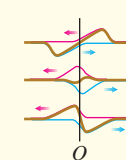
(inverse-square law for intensity)



Wave superposition: A wave reflects when it reaches a boundary of its medium. At any point where two or more waves overlap, the total displacement is the sum of the displacements of the individual waves (principle of superposition).

$$y(x, t) = y_1(x, t) + y_2(x, t) \quad (15.27)$$

(principle of superposition)



Standing waves on a string: When a sinusoidal wave is reflected from a fixed or free end of a stretched string, the incident and reflected waves combine to form a standing sinusoidal wave with nodes and antinodes. Adjacent nodes are spaced a distance $\lambda/2$ apart, as are adjacent antinodes. (See Example 15.6.)

When both ends of a string with length L are held fixed, standing waves can occur only when L is an integer multiple of $\lambda/2$. Each frequency with its associated vibration pattern is called a normal mode. (See Examples 15.7 and 15.8.)

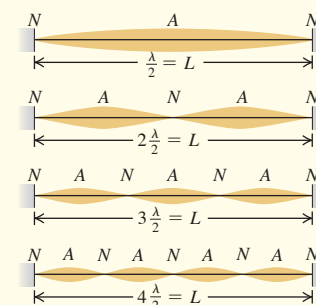
$$y(x, t) = (A_{SW} \sin kx) \sin \omega t \quad (15.28)$$

(standing wave on a string, fixed end at $x = 0$)

$$f_n = n \frac{v}{2L} = n f_1 \quad (n = 1, 2, 3, \dots) \quad (15.33)$$

$$f_1 = \frac{1}{2L} \sqrt{\frac{F}{\mu}} \quad (15.35)$$

(string fixed at both ends)





GUIDED PRACTICE

For assigned homework and other learning materials, go to Mastering Physics.

KEY EXAMPLE VARIATION PROBLEMS

Be sure to review **EXAMPLES 15.1, 15.2, and 15.3** (Sections 15.2, 15.3, and 15.4) before attempting these problems.

VP15.3.1 A boat is at anchor outside a harbor. A steady sinusoidal ocean wave makes the boat bob up and down with a period of 5.10 s and an amplitude of 1.00 m. The wave has wavelength 30.5 m. For this wave, what are (a) the frequency, (b) the wave speed, (c) the angular frequency, and (d) the wave number?

VP15.3.2 Sound waves in the thin Martian atmosphere travel at 245 m/s. (a) What are the period and wavelength of a 125 Hz sound wave in the Martian atmosphere? (b) What are the frequency and angular frequency of a sound wave in the Martian atmosphere that has wavelength 3.00 m?

VP15.3.3 You are testing a mountain climbing rope that has a linear mass density of 0.0650 kg/m. The rope is held horizontal and is under a tension of 8.00×10^2 N to simulate the stress of supporting a mountain climber's weight. (a) What is the speed of transverse waves on this rope? (b) You oscillate one end of the rope up and down in SHM with frequency 25.0 Hz and amplitude 5.00 mm. What is the wavelength of the resulting waves on the rope? (c) At $t = 0$ the end you are oscillating is at its maximum positive displacement and is instantaneously at rest. Write an equation for the displacement as a function of time at a point 2.50 m from that end. Assume that no wave bounces back from the other end.

VP15.3.4 The tension in a long string is 25.0 N. You oscillate one end of the string up and down with frequency 45.0 Hz. When this end is at its maximum upward displacement, the nearest point that is at its maximum negative displacement is 0.400 m down the string. What are (a) the speed of waves on the string and (b) the linear mass density of the string?

Be sure to review **EXAMPLES 15.4 and 15.5** (Section 15.5) before attempting these problems.

VP15.5.1 An athlete exerts a tension of 6.00×10^2 N on one end of a horizontal rope that has length 50.0 m and mass 2.50 kg. The other end is tied to a post. If she wiggles the rope with period 0.575 s and amplitude 3.00 cm, what are (a) the angular frequency of the oscillation and (b) the average rate at which energy is transferred along the rope?

VP15.5.2 A length of piano wire (mass density 5.55×10^{-4} kg/m) is under 185 N of tension. A sinusoidal wave of frequency 256 Hz carries a maximum power of 5.20 W along the wire. What is the amplitude of this wave?

VP15.5.3 A portable audio speaker has a power output of 8.00 W. (a) If the speaker emits sound equally in all directions, what is the sound intensity at a distance of 2.00 m from the speaker? (b) At what distance from the speaker is the intensity equal to 0.045 W/m^2 ?

VP15.5.4 The “ears” of a frog are two circular membranes located behind the frog’s eyes. In one species of frog each membrane is 0.500 cm in radius. If a source of sound has a power output of 2.50×10^{-6} W, emits sound equally in all directions, and is located 1.50 m from the frog, how much sound energy arrives at one of the membranes each second?

Be sure to review **EXAMPLES 15.6, 15.7, and 15.8** (Sections 15.7 and 15.8) before attempting these problems.

VP15.8.1 For a standing wave on a string, the distance between nodes is 0.125 m, the frequency is 256 Hz, and the amplitude is 1.40×10^{-3} m. What are (a) the speed of waves on this string, (b) the maximum transverse velocity at an antinode, and (c) the maximum transverse acceleration at an antinode?

VP15.8.2 The G string of a guitar has a fundamental frequency of 196 Hz. The linear mass density of the string is 2.29×10^{-3} kg/m, and the length of string that is free to vibrate (between the nut and bridge of the guitar) is 0.641 m. What are (a) the speed of waves on the G string and (b) the tension in this string?

VP15.8.3 A cable is stretched between two posts 3.00 m apart. The speed of waves on this cable is 96.0 m/s, and the tension in the cable is 175 N. If a standing wave on this cable has five antinodes, what are (a) the wavelength of the standing wave, (b) the frequency of the standing wave, and (c) the linear mass density of the cable?

VP15.8.4 One of the strings on a musical instrument is 0.500 m in length and has linear mass density 1.17×10^{-3} kg/m. The second harmonic on this string has frequency 512 Hz. (a) What is the tension in the string? (b) The speed of sound in air at 20°C is 344 m/s. If the string is vibrating at its fundamental frequency, what is the wavelength of the sound wave that the string produces in air?

BRIDGING PROBLEM Waves on a Rotating Rope

A uniform rope with length L and mass m is held at one end and whirled in a horizontal circle with angular velocity ω . You can ignore the force of gravity on the rope. (a) At a point on the rope a distance r from the end that is held, what is the tension F ? (b) What is the speed of transverse waves at this point? (c) Find the time required for a transverse wave to travel from one end of the rope to the other.

SOLUTION GUIDE

IDENTIFY and SET UP

1. Draw a sketch of the situation and label the distances r and L . The tension in the rope will be different at different values of r . Do you see why? Where on the rope do you expect the tension to be greatest? Where do you expect it will be least?

2. Where on the rope do you expect the wave speed to be greatest? Where do you expect it will be least?
3. Think about the portion of the rope that is farther out than r from the end that is held. What forces act on this portion? (Remember that you can ignore gravity.) What is the mass of this portion? How far is its center of mass from the rotation axis?
4. List the unknown quantities and decide which are your target variables.

EXECUTE

5. Draw a free-body diagram for the portion of the rope that is farther out than r from the end that is held.

6. Use your free-body diagram to help you determine the tension in the rope at distance r .
7. Use your result from step 6 to find the wave speed at distance r .
8. Use your result from step 7 to find the time for a wave to travel from one end to the other. (*Hint:* The wave speed is $v = dr/dt$, so the time for the wave to travel a distance dr along the rope is $dt = dr/v$. Integrate this to find the total time. See Appendix B.)

EVALUATE

9. Do your results for parts (a) and (b) agree with your expectations from steps 1 and 2? Are the units correct?
10. Check your result for part (a) by considering the net force on a small segment of the rope at distance r with length dr and mass $dm = (m/L)dr$. [*Hint:* The tension forces on this segment are $F(r)$ on one side and $F(r + dr)$ on the other side. You will get an equation for dF/dr that you can integrate to find F as a function of r .]

PROBLEMS

•, ••, •••: Difficulty levels. **CP**: Cumulative problems incorporating material from earlier chapters. **CALC**: Problems requiring calculus. **DATA**: Problems involving real data, scientific evidence, experimental design, and/or statistical reasoning. **BIO**: Biosciences problems.

DISCUSSION QUESTIONS

Q15.1 Two waves travel on the same string. Is it possible for them to have (a) different frequencies; (b) different wavelengths; (c) different speeds; (d) different amplitudes; (e) the same frequency but different wavelengths? Explain your reasoning.

Q15.2 Under a tension F , it takes 2.00 s for a pulse to travel the length of a taut wire. What tension is required (in terms of F) for the pulse to take 6.00 s instead? Explain how you arrive at your answer.

Q15.3 What kinds of energy are associated with waves on a stretched string? How could you detect such energy experimentally?

Q15.4 The amplitude of a wave decreases gradually as the wave travels along a long, stretched string. What happens to the energy of the wave when this happens?

Q15.5 For the wave motions discussed in this chapter, does the speed of propagation depend on the amplitude? What makes you say this?

Q15.6 The speed of ocean waves depends on the depth of the water; the deeper the water, the faster the wave travels. Use this to explain why ocean waves crest and “break” as they near the shore.

Q15.7 Is it possible to have a longitudinal wave on a stretched string? Why or why not? Is it possible to have a transverse wave on a steel rod? Again, why or why not? If your answer is yes in either case, explain how you would create such a wave.

Q15.8 For transverse waves on a string, is the wave speed the same as the speed of any part of the string? Explain the difference between these two speeds. Which one is constant?

Q15.9 The four strings on a violin have different thicknesses, but are all under approximately the same tension. Do waves travel faster on the thick strings or the thin strings? Why? How does the fundamental vibration frequency compare for the thick versus the thin strings?

Q15.10 A sinusoidal wave can be described by a cosine function, which is negative just as often as positive. So why isn't the average power delivered by this wave zero?

Q15.11 Two strings of different mass per unit length μ_1 and μ_2 are tied together and stretched with a tension F . A wave travels along the string and passes the discontinuity in μ . Which of the following wave properties will be the same on both sides of the discontinuity, and which will change: speed of the wave; frequency; wavelength? Explain the physical reasoning behind each answer.

Q15.12 A long rope with mass m is suspended from the ceiling and hangs vertically. A wave pulse is produced at the lower end of the rope, and the pulse travels up the rope. Does the speed of the wave pulse change as it moves up the rope, and if so, does it increase or decrease? Explain.

Q15.13 In a transverse wave on a string, the motion of the string is perpendicular to the length of the string. How, then, is it possible for energy to move along the length of the string?

Q15.14 Energy can be transferred along a string by wave motion. However, in a standing wave on a string, no energy can ever be transferred past a node. Why not?

Q15.15 Can a standing wave be produced on a string by superposing two waves traveling in opposite directions with the same frequency but different amplitudes? Why or why not? Can a standing wave be produced by superposing two waves traveling in opposite directions with different frequencies but the same amplitude? Why or why not?

Q15.16 If you stretch a rubber band and pluck it, you hear a (somewhat) musical tone. How does the frequency of this tone change as you stretch the rubber band further? (Try it!) Does this agree with Eq. (15.35) for a string fixed at both ends? Explain.

Q15.17 A musical interval of an *octave* corresponds to a factor of 2 in frequency. By what factor must the tension in a guitar or violin string be increased to raise its pitch one octave? To raise it two octaves? Explain your reasoning. Is there any danger in attempting these changes in pitch?

Q15.18 By touching a string lightly at its center while bowing, a violinist can produce a note exactly one octave above the note to which the string is tuned—that is, a note with exactly twice the frequency. Why is this possible?

Q15.19 As we discussed in Section 15.1, water waves are a combination of longitudinal and transverse waves. Defend the following statement: “When water waves hit a vertical wall, the wall is a node of the longitudinal displacement but an antinode of the transverse displacement.”

Q15.20 Violins are short instruments, while cellos and basses are long. In terms of the frequency of the waves they produce, explain why this is so.

Q15.21 What is the purpose of the frets on a guitar? In terms of the frequency of the vibration of the strings, explain their use.

EXERCISES

Section 15.2 Periodic Waves

15.1 • The speed of sound in air at 20°C is 344 m/s. (a) What is the wavelength of a sound wave with a frequency of 784 Hz, corresponding to the note G₅ on a piano, and how many milliseconds does each vibration take? (b) What is the wavelength of a sound wave one octave higher (twice the frequency) than the note in part (a)?

15.2 • BIO Ultrasound Imaging. Sound having frequencies above the range of human hearing (about 20,000 Hz) is called *ultrasound*. Waves above this frequency can be used to penetrate the body and to produce images by reflecting from surfaces. In a typical ultrasound scan, the waves travel through body tissue with a speed of 1500 m/s. For a good, detailed image, the wavelength should be no more than 1.0 mm. What frequency sound is required for a good scan?

15.3 • Tsunami! On December 26, 2004, a great earthquake occurred off the coast of Sumatra and triggered immense waves (tsunami) that killed more than 200,000 people. Satellites observing these waves from space measured 800 km from one wave crest to the next and a period between waves of 1.0 hour. What was the speed of these waves in m/s and in km/h? Does your answer help you understand why the waves caused such devastation?

15.4 •• A fisherman notices that his boat is moving up and down periodically, owing to waves on the surface of the water. It takes 2.5 s for the boat to travel from its highest point to its lowest, a total distance of 0.53 m. The fisherman sees that the wave crests are spaced 4.8 m apart. (a) How fast are the waves traveling? (b) What is the amplitude of each wave? (c) If the total vertical distance traveled by the boat were 0.30 m but the other data remained the same, how would the answers to parts (a) and (b) change?

15.5 • BIO (a) Audible wavelengths. The range of audible frequencies is from about 20 Hz to 20,000 Hz. What is the range of the wavelengths of audible sound in air? (b) **Visible light.** The range of visible light extends from 380 nm to 750 nm. What is the range of visible frequencies of light? (c) **Brain surgery.** Surgeons can remove brain tumors by using a cavitron ultrasonic surgical aspirator, which produces sound waves of frequency 23 kHz. What is the wavelength of these waves in air? (d) **Sound in the body.** What would be the wavelength of the sound in part (c) in bodily fluids in which the speed of sound is 1480 m/s but the frequency is unchanged?

Section 15.3 Mathematical Description of a Wave

15.6 • A small bead of mass 4.00 g is attached to a horizontal string. Transverse waves of amplitude $A = 0.800$ cm and frequency $f = 20.0$ Hz are set up on the string. Assume the mass of the bead is small enough that the bead doesn't alter the wave motion. During the wave motion, what is the maximum vertical force that the string exerts on the bead?

15.7 • Transverse waves on a string have wave speed 8.00 m/s, amplitude 0.0700 m, and wavelength 0.320 m. The waves travel in the $-x$ -direction, and at $t = 0$ the $x = 0$ end of the string has its maximum upward displacement. (a) Find the frequency, period, and wave number of these waves. (b) Write a wave function describing the wave. (c) Find the transverse displacement of a particle at $x = 0.360$ m at time $t = 0.150$ s. (d) How much time must elapse from the instant in part (c) until the particle at $x = 0.360$ m next has maximum upward displacement?

15.8 • A certain transverse wave is described by

$$y(x, t) = (6.50 \text{ mm}) \cos 2\pi \left(\frac{x}{28.0 \text{ cm}} - \frac{t}{0.0360 \text{ s}} \right)$$

Determine the wave's (a) amplitude; (b) wavelength; (c) frequency; (d) speed of propagation; (e) direction of propagation.

15.9 • CALC Which of the following wave functions satisfies the wave equation, Eq. (15.12)? (a) $y(x, t) = A \cos(kx + \omega t)$; (b) $y(x, t) = A \sin(kx + \omega t)$; (c) $y(x, t) = A(\cos kx + \cos \omega t)$. (d) For the wave of part (b), write the equations for the transverse velocity and transverse acceleration of a particle at point x .

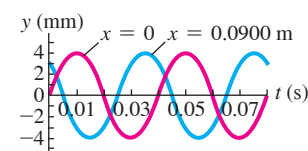
15.10 • A water wave traveling in a straight line on a lake is described by the equation

$$y(x, t) = (2.75 \text{ cm}) \cos(0.410 \text{ rad/cm } x + 6.20 \text{ rad/s } t)$$

where y is the displacement perpendicular to the undisturbed surface of the lake. (a) How much time does it take for one complete wave pattern to go past a fisherman in a boat at anchor, and what horizontal distance does the wave crest travel in that time? (b) What are the wave number and the number of waves per second that pass the fisherman? (c) How fast does a wave crest travel past the fisherman, and what is the maximum speed of his cork floater as the wave causes it to bob up and down?

15.11 • A sinusoidal wave is propagating along a stretched string that lies along the x -axis. The displacement of the string as a function of time is graphed in Fig. E15.11 for particles at $x = 0$ and at $x = 0.0900$ m. (a) What is the amplitude of the wave? (b) What is the period of

Figure E15.11



the wave? (c) You are told that the two points $x = 0$ and $x = 0.0900$ m are within one wavelength of each other. If the wave is moving in the $+x$ -direction, determine the wavelength and the wave speed. (d) If instead the wave is moving in the $-x$ -direction, determine the wavelength and the wave speed. (e) Would it be possible to determine definitively the wavelengths in parts (c) and (d) if you were not told that the two points were within one wavelength of each other? Why or why not?

15.12 •• CALC Speed of Propagation vs. Particle Speed. (a) Show that Eq. (15.3) may be written as

$$y(x, t) = A \cos \left[\frac{2\pi}{\lambda} (x - vt) \right]$$

(b) Use $y(x, t)$ to find an expression for the transverse velocity v_y of a particle in the string on which the wave travels. (c) Find the maximum speed of a particle of the string. Under what circumstances is this equal to the propagation speed v ? Less than v ? Greater than v ?

15.13 •• A transverse wave on a string has amplitude 0.300 cm, wavelength 12.0 cm, and speed 6.00 cm/s. It is represented by $y(x, t)$ as given in Exercise 15.12. (a) At time $t = 0$, compute y at 1.5 cm intervals of x (that is, at $x = 0, x = 1.5 \text{ cm}, x = 3.0 \text{ cm}$, and so on) from $x = 0$ to $x = 12.0$ cm. Graph the results. This is the shape of the string at time $t = 0$. (b) Repeat the calculations for the same values of x at times $t = 0.400$ s and $t = 0.800$ s. Graph the shape of the string at these instants. In what direction is the wave traveling?

Section 15.4 Speed of a Transverse Wave

15.14 • A musical novice learns that doubling the fundamental increases the pitch by one octave and decides to do that to a guitar string. What factor increase in tension would be necessary? Do you think this would be a good idea?

15.15 • One end of a horizontal rope is attached to a prong of an electrically driven tuning fork that vibrates the rope transversely at 120 Hz. The other end passes over a pulley and supports a 1.50 kg mass. The linear mass density of the rope is 0.0480 kg/m. (a) What is the speed of a transverse wave on the rope? (b) What is the wavelength? (c) How would your answers to parts (a) and (b) change if the mass were increased to 3.00 kg?

15.16 • With what tension must a rope with length 2.50 m and mass 0.120 kg be stretched for transverse waves of frequency 40.0 Hz to have a wavelength of 0.750 m?

15.17 •• The upper end of a 3.8-m-long steel wire is fastened to the ceiling, and a 54 kg object is suspended from the lower end of the wire. You observe that it takes a transverse pulse 0.049 s to travel from the bottom to the top of the wire. What is the mass of the wire?

15.18 •• A 1.50 m string of weight 0.0125 N is tied to the ceiling at its upper end, and the lower end supports a weight W . Ignore the very small variation in tension along the length of the string that is produced by the weight of the string. When you pluck the string slightly, the waves traveling up the string obey the equation

$$y(x, t) = (8.50 \text{ mm})\cos(172 \text{ rad/m } x - 4830 \text{ rad/s } t)$$

Assume that the tension of the string is constant and equal to W . (a) How much time does it take a pulse to travel the full length of the string? (b) What is the weight W ? (c) How many wavelengths are on the string at any instant of time? (d) What is the equation for waves traveling *down* the string?

15.19 • A thin, 75.0 cm wire has a mass of 16.5 g. One end is tied to a nail, and the other end is attached to a screw that can be adjusted to vary the tension in the wire. (a) To what tension (in newtons) must you adjust the screw so that a transverse wave of wavelength 3.33 cm makes 625 vibrations per second? (b) How fast would this wave travel?

15.20 •• A heavy rope 6.00 m long and weighing 29.4 N is attached at one end to a ceiling and hangs vertically. A 0.500 kg mass is suspended from the lower end of the rope. What is the speed of transverse waves on the rope at the (a) bottom of the rope, (b) middle of the rope, and (c) top of the rope? (d) Is the tension in the middle of the rope the average of the tensions at the top and bottom of the rope? Is the wave speed at the middle of the rope the average of the wave speeds at the top and bottom? Explain.

Section 15.5 Energy in Wave Motion

15.21 •• In Example 15.4 the average power that Throckmorton puts into the clothesline is small, about 1 W. How much better can you do? Assume a rope that has linear mass density 0.500 kg/m, so 1 m of the rope weighs about 1 lb. The rope is long and is attached to a post at one end. You hold the other end of the rope in your hand and supply sinusoidal wave pulses by moving your arm up and down. Estimate the amplitude of the pulses to be the length of your arm. Estimate the maximum tension you can supply to the rope by pulling on it horizontally, and estimate the time for you to complete each pulse. (a) Ignoring any effects from reflection of the pulses from the other end of the rope, what average power can you supply to the rope? (b) You try to increase your power output by halving the amplitude so you can double the frequency of the pulses. What change in P_{av} does this produce?

15.22 •• A piano wire with mass 3.00 g and length 80.0 cm is stretched with a tension of 25.0 N. A wave with frequency 120.0 Hz and amplitude 1.6 mm travels along the wire. (a) Calculate the average power carried by the wave. (b) What happens to the average power if the wave amplitude is halved?

15.23 • A horizontal wire is stretched with a tension of 94.0 N, and the speed of transverse waves for the wire is 406 m/s. What must the amplitude of a traveling wave of frequency 69.0 Hz be for the average power carried by the wave to be 0.365 W?

15.24 •• Threshold of Pain. You are investigating the report of a UFO landing in an isolated portion of New Mexico, and you encounter a strange object that is radiating sound waves uniformly in all directions. Assume that the sound comes from a point source and that you can ignore reflections. You are slowly walking toward the source. When you are 7.5 m from it, you measure its intensity to be 0.11 W/m^2 . An intensity of 1.0 W/m^2 is often used as the “threshold of pain.” How much closer to the source can you move before the sound intensity reaches this threshold?

15.25 •• A jet plane at takeoff can produce sound of intensity 10.0 W/m^2 at 30.0 m away. But you prefer the tranquil sound of normal conversation, which is $1.0 \mu\text{W/m}^2$. Assume that the plane behaves like a point source of sound. (a) What is the closest distance you should live from the airport runway to preserve your peace of mind? (b) What intensity from the jet does your friend experience if she lives twice as far from the runway as you do? (c) What power of sound does the jet produce at takeoff?

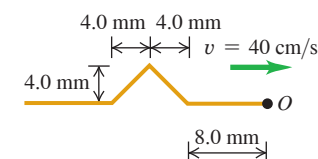
15.26 • A fellow student with a mathematical bent tells you that the wave function of a traveling wave on a thin rope is $y(x, t) = (2.30 \text{ mm})\cos[(6.98 \text{ rad/m})x + (742 \text{ rad/s})t]$. Being more practical, you measure the rope to have a length of 1.35 m and a mass of 0.00338 kg. You are then asked to determine the following: (a) amplitude; (b) frequency; (c) wavelength; (d) wave speed; (e) direction the wave is traveling; (f) tension in the rope; (g) average power transmitted by the wave.

15.27 • Energy Output. By measurement you determine that sound waves are spreading out equally in all directions from a point source and that the intensity is 0.026 W/m^2 at a distance of 4.3 m from the source. (a) What is the intensity at a distance of 3.1 m from the source? (b) How much sound energy does the source emit in one hour if its power output remains constant?

Section 15.6 Wave Interference, Boundary Conditions, and Superposition

15.28 • Reflection. A wave pulse on a string has the dimensions shown in **Fig. E15.28** at $t = 0$. The wave speed is 40 cm/s. (a) If point O is a fixed end, draw the total wave on the string at $t = 15 \text{ ms}$, 20 ms , 25 ms , 30 ms , 35 ms , 40 ms , and 45 ms .

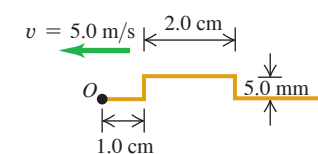
Figure E15.28



(b) Repeat part (a) for the case in which point O is a free end.

15.29 • Reflection. A wave pulse on a string has the dimensions shown in **Fig. E15.29** at $t = 0$. The wave speed is 5.0 m/s. (a) If point O is a fixed end, draw the total wave on the string at $t = 1.0 \text{ ms}$, 2.0 ms , 3.0 ms , 4.0 ms , 5.0 ms , 6.0 ms , and 7.0 ms .

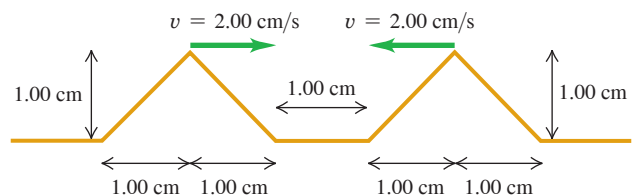
Figure E15.29



(b) Repeat part (a) for the case in which point O is a free end.

15.30 • Interference of Triangular Pulses. Two triangular wave pulses are traveling toward each other on a stretched string as shown in **Fig. E15.30**. Each pulse is identical to the other and travels at 2.00 cm/s. The leading edges of the pulses are 1.00 cm apart at $t = 0$. Sketch the shape of the string at $t = 0.250 \text{ s}$, $t = 0.500 \text{ s}$, $t = 0.750 \text{ s}$, $t = 1.000 \text{ s}$, and $t = 1.250 \text{ s}$.

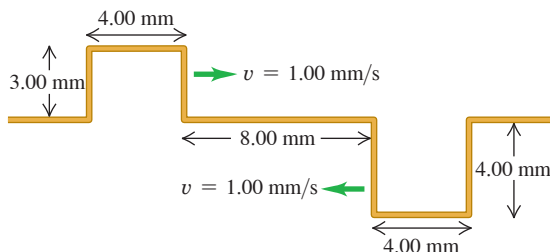
Figure E15.30



15.31 • Suppose that the left-traveling pulse in Exercise 15.30 is *below* the level of the unstretched string instead of above it. Make the same sketches that you did in that exercise.

15.32 •• Interference of Rectangular Pulses. Figure E15.32 shows two rectangular wave pulses on a stretched string traveling toward each other. Each pulse is traveling with a speed of 1.00 mm/s and has the height and width shown in the figure. If the leading edges of the pulses are 8.00 mm apart at $t = 0$, sketch the shape of the string at $t = 4.00$ s, $t = 6.00$ s, and $t = 10.0$ s.

Figure E15.32



Section 15.7 Standing Waves on a String

Section 15.8 Normal Modes of a String

15.33 •• For a violin, estimate the length of the portions of the strings that are free to vibrate. (a) The frequency of the note played by the open E5 string vibrating in its fundamental standing wave is 659 Hz. Use your estimate of the length to calculate the wave speed for the transverse waves on the string. (b) The vibrating string produces sound waves in air with the same frequency as that of the string. Use 344 m/s for the speed of sound in air and calculate the wavelength of the E5 note in air. Which is larger: the wavelength on the string or the wavelength in air? (c) Repeat parts (a) and (b) for a bass viol, which is typically played by a person standing up. Start your calculation by estimating the length of the bass viol string that is free to vibrate. The G2 string produces a note with frequency 98 Hz when vibrating in its fundamental standing wave.

15.34 •• CALC Adjacent antinodes of a standing wave on a string are 15.0 cm apart. A particle at an antinode oscillates in simple harmonic motion with amplitude 0.850 cm and period 0.0750 s. The string lies along the $+x$ -axis and is fixed at $x = 0$. (a) How far apart are the adjacent nodes? (b) What are the wavelength, amplitude, and speed of the two traveling waves that form this pattern? (c) Find the maximum and minimum transverse speeds of a point at an antinode. (d) What is the shortest distance along the string between a node and an antinode?

15.35 • Standing waves on a wire are described by Eq. (15.28), with $A_{\text{sw}} = 2.50$ mm, $\omega = 942$ rad/s, and $k = 0.750\pi$ rad/m. The left end of the wire is at $x = 0$. At what distances from the left end are (a) the nodes of the standing wave and (b) the antinodes of the standing wave?

15.36 • A 1.50-m-long rope is stretched between two supports with a tension that makes the speed of transverse waves 62.0 m/s. What are the wavelength and frequency of (a) the fundamental; (b) the second overtone; (c) the fourth harmonic?

15.37 • A wire with mass 40.0 g is stretched so that its ends are tied down at points 80.0 cm apart. The wire vibrates in its fundamental mode with frequency 60.0 Hz and with an amplitude at the antinodes of 0.300 cm. (a) What is the speed of propagation of transverse waves in the wire? (b) Compute the tension in the wire. (c) Find the maximum transverse velocity and acceleration of particles in the wire.

15.38 • A piano tuner stretches a steel piano wire with a tension of 800 N. The steel wire is 0.400 m long and has a mass of 3.00 g. (a) What is the frequency of its fundamental mode of vibration? (b) What is the number of the highest harmonic that could be heard by a person who is capable of hearing frequencies up to 10,000 Hz?

15.39 • CALC A thin, taut string tied at both ends and oscillating in its third harmonic has its shape described by the equation $y(x, t) = (5.60 \text{ cm}) \sin[(0.0340 \text{ rad/cm})x] \sin[(50.0 \text{ rad/s})t]$, where the origin is at the left end of the string, the x -axis is along the string, and the y -axis is perpendicular to the string. (a) Draw a sketch that shows the standing-wave pattern. (b) Find the amplitude of the two traveling waves that make up this standing wave. (c) What is the length of the string? (d) Find the wavelength, frequency, period, and speed of the traveling waves. (e) Find the maximum transverse speed of a point on the string. (f) What would be the equation $y(x, t)$ for this string if it were vibrating in its eighth harmonic?

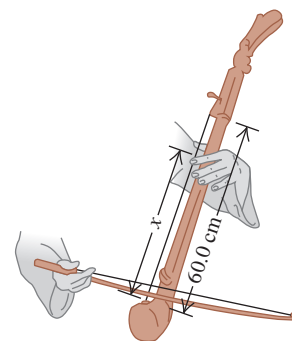
15.40 • The wave function of a standing wave is $y(x, t) = (4.44 \text{ mm}) \sin[(32.5 \text{ rad/m})x] \sin[(754 \text{ rad/s})t]$. For the two traveling waves that make up this standing wave, find the (a) amplitude; (b) wavelength; (c) frequency; (d) wave speed; (e) wave functions. (f) From the information given, can you determine which harmonic this is? Explain.

15.41 • Standing waves are produced on a string that is held fixed at both ends. The tension in the string is kept constant. (a) For the second overtone standing wave the node-to-node distance is 8.00 cm. What is the length of the string? (b) What is the node-to-node distance for the fourth harmonic standing wave?

15.42 •• One string of a certain musical instrument is 75.0 cm long and has a mass of 8.75 g. It is being played in a room where the speed of sound is 344 m/s. (a) To what tension must you adjust the string so that, when vibrating in its second overtone, it produces sound of wavelength 0.765 m? (Assume that the breaking stress of the wire is very large and isn't exceeded.) (b) What frequency sound does this string produce in its fundamental mode of vibration?

15.43 • The portion of the string of a certain musical instrument between the bridge and upper end of the finger board (that part of the string that is free to vibrate) is 60.0 cm long, and this length of the string has mass 2.00 g. The string sounds an A_4 note (440 Hz) when played. (a) Where must the player put a finger (what distance x from the bridge) to play a D_5 note (587 Hz)? (See Fig. E15.43.) For both the A_4 and D_5 notes, the string vibrates in its fundamental mode. (b) Without retuning, is it possible to play a G_4 note (392 Hz) on this string? Why or why not?

Figure E15.43



15.44 • (a) A horizontal string tied at both ends is vibrating in its fundamental mode. The traveling waves have speed v , frequency f , amplitude A , and wavelength λ . Calculate the maximum transverse velocity and maximum transverse acceleration of points located at (i) $x = \lambda/2$, (ii) $x = \lambda/4$, and (iii) $x = \lambda/8$, from the left-hand end of the string. (b) At each of the points in part (a), what is the amplitude of the motion? (c) At each of the points in part (a), how much time does it take the string to go from its largest upward displacement to its largest downward displacement?

PROBLEMS

15.45 •• A sinusoidal wave with wavelength 0.400 m travels along a string. The maximum transverse speed of a point on the string is 3.00 m/s and the maximum transverse acceleration is $8.50 \times 10^4 \text{ m/s}^2$. What are the propagation speed v and the amplitude A of the wave?

15.46 • A transverse wave on a rope is given by

$$y(x, t) = (0.750 \text{ cm}) \cos \pi[(0.400 \text{ cm}^{-1})x + (250 \text{ s}^{-1})t]$$

(a) Find the amplitude, period, frequency, wavelength, and speed of propagation. (b) Sketch the shape of the rope at these values of t : 0, 0.0005 s, 0.0010 s. (c) Is the wave traveling in the $+x$ - or $-x$ -direction? (d) The mass per unit length of the rope is 0.0500 kg/m. Find the tension. (e) Find the average power of this wave.

15.47 • CALC A transverse sine wave with an amplitude of 2.50 mm and a wavelength of 1.80 m travels from left to right along a long, horizontal, stretched string with a speed of 36.0 m/s. Take the origin at the left end of the undisturbed string. At time $t = 0$ the left end of the string has its maximum upward displacement. (a) What are the frequency, angular frequency, and wave number of the wave? (b) What is the function $y(x, t)$ that describes the wave? (c) What is $y(t)$ for a particle at the left end of the string? (d) What is $y(t)$ for a particle 1.35 m to the right of the origin? (e) What is the maximum magnitude of transverse velocity of any particle of the string? (f) Find the transverse displacement and the transverse velocity of a particle 1.35 m to the right of the origin at time $t = 0.0625$ s.

15.48 •• CP A 1750 N irregular beam is hanging horizontally by its ends from the ceiling by two vertical wires (A and B), each 1.25 m long and weighing 0.290 N. The center of gravity of this beam is one-third of the way along the beam from the end where wire A is attached. If you pluck both strings at the same time at the beam, what is the time delay between the arrival of the two pulses at the ceiling? Which pulse arrives first? (Ignore the effect of the weight of the wires on the tension in the wires.)

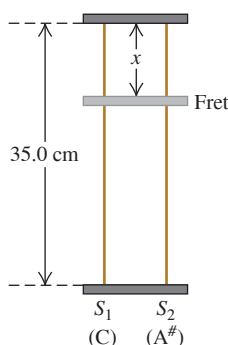
15.49 ••• CP One end of a light uniform rod is attached to a wall by a frictionless hinge. The rod is held in a horizontal position by a wire that runs from the other end of the rod to the wall. The wire has length 2.00 m and makes an angle of 30.0° with the rod. A block with mass m is suspended by a light rope attached to the middle of the rod. The transverse fundamental standing wave on the wire has frequency f . The mass m is varied and for each value the frequency is measured. You plot f^2 versus m and find that your data lie close to a straight line with slope $20.4 \text{ kg}^{-1} \cdot \text{s}^{-2}$. What is the mass of the wire? Assume that the change in the length of the wire when the tension changes is small enough to neglect.

15.50 •• Weightless Ant. An ant with mass m is standing peacefully on top of a horizontal, stretched rope. The rope has mass per unit length μ and is under tension F . Without warning, Cousin Throckmorton starts a sinusoidal transverse wave of wavelength λ propagating along the rope. The motion of the rope is in a vertical plane. What minimum wave amplitude will make the ant become momentarily weightless? Assume that m is so small that the presence of the ant has no effect on the propagation of the wave.

15.51 •• You must determine the length of a long, thin wire that is suspended from the ceiling in the atrium of a tall building. A 2.00-cm-long piece of the wire is left over from its installation. Using an analytical balance, you determine that the mass of the spare piece is $14.5 \mu\text{g}$. You then hang a 0.400 kg mass from the lower end of the long, suspended wire. When a small-amplitude transverse wave pulse is sent up that wire, sensors at both ends measure that it takes the wave pulse 26.7 ms to travel the length of the wire. (a) Use these measurements to calculate the length of the wire. Assume that the weight of the wire has a negligible effect on the speed of the transverse waves. (b) Discuss the accuracy of the approximation made in part (a).

15.52 •• Music. You are designing a two-string instrument with metal strings 35.0 cm long, as shown in **Fig. P15.52**. Both strings are under the same tension. String S_1 has a mass of 8.00 g and produces the note middle C (frequency 262 Hz) in its fundamental mode. (a) What should be the tension in the string? (b) What should be the mass of string S_2 so that it will produce A-sharp (frequency 466 Hz) as its fundamental? (c) To extend the range of your instrument, you include

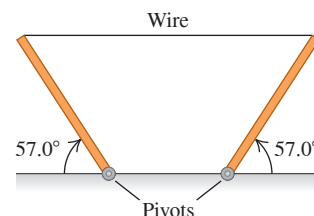
Figure P15.52



a fret located just under the strings but not normally touching them. How far from the upper end should you put this fret so that when you press S_1 tightly against it, this string will produce C-sharp (frequency 277 Hz) in its fundamental? That is, what is x in the figure? (d) If you press S_2 against the fret, what frequency of sound will it produce in its fundamental?

15.53 ••• CP A 5.00 m, 0.732 kg wire is used to support two uniform 235 N posts of equal length (**Fig. P15.53**). Assume that the wire is essentially horizontal and that the speed of sound is 344 m/s. A strong wind is blowing, causing the wire to vibrate in its 5th overtone. What are the frequency and wavelength of the sound this wire produces?

Figure P15.53



15.54 ••• CP You are exploring a newly discovered planet. The radius of the planet is 7.20×10^7 m. You suspend a lead weight from the lower end of a light string that is 4.00 m long and has mass 0.0280 kg. You measure that it takes 0.0685 s for a transverse pulse to travel from the lower end to the upper end of the string. On the earth, for the same string and lead weight, it takes 0.0390 s for a transverse pulse to travel the length of the string. The weight of the string is small enough that you ignore its effect on the tension in the string. Assuming that the mass of the planet is distributed with spherical symmetry, what is its mass?

15.55 •• For a string stretched between two supports, two successive standing-wave frequencies are 525 Hz and 630 Hz. There are other standing-wave frequencies lower than 525 Hz and higher than 630 Hz. If the speed of transverse waves on the string is 384 m/s, what is the length of the string? Assume that the mass of the wire is small enough for its effect on the tension in the wire to be ignored.

15.56 •• Transverse standing waves are produced on a string that has length 0.800 m and is held fixed at each end. Each standing-wave pattern has a node at the fixed ends plus additional nodes along the length of the string. You measure the frequencies f_n for standing waves that have n of these nodes along their length. The tension in the string is kept constant. You plot n versus f_n and find that your data lie close to a straight line that has slope 7.30×10^{-3} s. What is the speed of transverse waves on the string?

15.57 ••• CP A 1.80-m-long uniform bar that weighs 638 N is suspended in a horizontal position by two vertical wires that are attached to the ceiling. One wire is aluminum and the other is copper. The aluminum wire is attached to the left-hand end of the bar, and the copper wire is attached 0.40 m to the left of the right-hand end. Each wire has length 0.600 m and a circular cross section with radius 0.280 mm. What is the fundamental frequency of transverse standing waves for each wire?

15.58 •• CALC A transverse standing wave is set up on a string that is held fixed at both ends. The amplitude of the standing wave at an antinode is 1.80 mm and the speed of propagation of transverse waves on the string is 260 m/s. The string extends along the x -axis, with one of the fixed ends at $x = 0$, so that there is a node at $x = 0$. The smallest value of x where there is an antinode is $x = 0.150$ m. (a) What is the maximum transverse speed of a point on the string at an antinode? (b) What is the maximum transverse speed of a point on the string at $x = 0.075$ m?

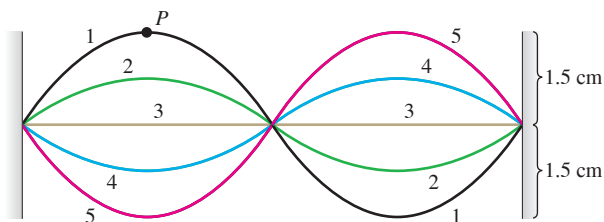
15.59 •• A horizontal wire is tied to supports at each end and vibrates in its second-overtone standing wave. The tension in the wire is 5.00 N, and the node-to-node distance in the standing wave is 6.28 cm. (a) What is the length of the wire? (b) A point at an antinode of the standing wave on the wire travels from its maximum upward displacement to its maximum downward displacement in 8.40 ms. What is the wire's mass?

15.60 •• CP A vertical, 1.20 m length of 18 gauge (diameter of 1.024 mm) copper wire has a 100.0 N ball hanging from it. (a) What is the wavelength of the third harmonic for this wire? (b) A 500.0 N ball now *replaces* the original ball. What is the change in the wavelength of the third harmonic caused by replacing the light ball with the heavy one? (*Hint:* See Table 11.1 for Young's modulus.)

15.61 •• A sinusoidal transverse wave travels on a string. The string has length 8.00 m and mass 6.00 g. The wave speed is 30.0 m/s, and the wavelength is 0.200 m. (a) If the wave is to have an average power of 50.0 W, what must be the amplitude of the wave? (b) For this same string, if the amplitude and wavelength are the same as in part (a), what is the average power for the wave if the tension is increased such that the wave speed is doubled?

15.62 •• A vibrating string 50.0 cm long is under a tension of 1.00 N. The results from five successive stroboscopic pictures are shown in **Fig. P15.62**. The strobe rate is set at 5000 flashes per minute, and observations reveal that the maximum displacement occurred at flashes 1 and 5 with no other maxima in between. (a) Find the period, frequency, and wavelength for the traveling waves on this string. (b) In what normal mode (harmonic) is the string vibrating? (c) What is the speed of the traveling waves on the string? (d) How fast is point *P* moving when the string is in (i) position 1 and (ii) position 3? (e) What is the mass of this string?

Figure P15.62



15.63 •• CP A 1.005 m chain consists of small spherical beads, each with a mass of 1.00 g and a diameter of 5.00 mm, threaded on an elastic strand with negligible mass such that adjacent beads are separated by a center-to-center distance of 10.0 mm. There are beads at each end of the chain. The strand has a spring constant of 28.8 N/m. The chain is stretched horizontally on a frictionless tabletop to a length of 1.50 m, and the beads at both ends are fixed in place. (a) What is the linear mass density of the chain? (b) What is the tension in the chain? (c) With what speed would a pulse travel down the chain? (d) The chain is set vibrating and exhibits a standing-wave pattern with four antinodes. What is the frequency of this motion? (e) If the beads are numbered sequentially from 1 to 101, what are the numbers of the five beads that remain motionless? (f) The 13th bead has a maximum speed of 7.54 m/s. What is the amplitude of that bead's motion? (g) If $x_0 = 0$ corresponds to the center of the 1st bead and $x_{101} = 1.50$ m corresponds to the center of the 101st bead, what is the position x_n of the n th bead? (h) What is the maximum speed of the 30th bead?

15.64 •• A strong string of mass 3.00 g and length 2.20 m is tied to supports at each end and is vibrating in its fundamental mode. The maximum transverse speed of a point at the middle of the string is 9.00 m/s. The tension in the string is 330 N. (a) What is the amplitude of the standing wave at its antinode? (b) What is the magnitude of the maximum transverse acceleration of a point at the antinode?

15.65 •• A thin string 2.50 m in length is stretched with a tension of 90.0 N between two supports. When the string vibrates in its first overtone, a point at an antinode of the standing wave on the string has an amplitude of 3.50 cm and a maximum transverse speed of 28.0 m/s. (a) What is the string's mass? (b) What is the magnitude of the maximum transverse acceleration of this point on the string?

15.66 •• CALC A guitar string is vibrating in its fundamental mode, with nodes at each end. The length of the segment of the string that is free to vibrate is 0.386 m. The maximum transverse acceleration of a point at the middle of the segment is $8.40 \times 10^3 \text{ m/s}^2$ and the maximum transverse velocity is 3.80 m/s. (a) What is the amplitude of this standing wave? (b) What is the wave speed for the transverse traveling waves on this string?

15.67 •• A uniform cylindrical steel wire, 55.0 cm long and 1.14 mm in diameter, is fixed at both ends. To what tension must it be adjusted so that, when vibrating in its first overtone, it produces the note D-sharp of frequency 311 Hz? Assume that it stretches an insignificant amount. (*Hint:* See Table 12.1.)

15.68 •• A string with both ends held fixed is vibrating in its third harmonic. The waves have a speed of 192 m/s and a frequency of 240 Hz. The amplitude of the standing wave at an antinode is 0.400 cm. (a) Calculate the amplitude at points on the string a distance of (i) 40.0 cm; (ii) 20.0 cm; and (iii) 10.0 cm from the left end of the string. (b) At each point in part (a), how much time does it take the string to go from its largest upward displacement to its largest downward displacement? (c) Calculate the maximum transverse velocity and the maximum transverse acceleration of the string at each of the points in part (a).

15.69 •• CP A large rock that weighs 164.0 N is suspended from the lower end of a thin wire that is 3.00 m long. The density of the rock is 3200 kg/m^3 . The mass of the wire is small enough that its effect on the tension in the wire can be ignored. The upper end of the wire is held fixed. When the rock is in air, the fundamental frequency for transverse standing waves on the wire is 42.0 Hz. When the rock is totally submerged in a liquid, with the top of the rock just below the surface, the fundamental frequency for the wire is 28.0 Hz. What is the density of the liquid?

15.70 •• (a) Estimate the tension you would need to apply to a standard small rubber band to stretch it between your fingers to a doubled length of 10 cm. (b) Such a rubber band has a typical mass of 0.10 g. Use this value to estimate the mass density of the stretched rubber band in SI units. (c) Use your estimated values to determine the expected frequency of vibration when the string is plucked. (d) Is this result realistic?

15.71 •• Tuning an Instrument. A musician tunes the C-string of her instrument to a fundamental frequency of 65.4 Hz. The vibrating portion of the string is 0.600 m long and has a mass of 14.4 g. (a) With what tension must the musician stretch it? (b) What percent increase in tension is needed to increase the frequency from 65.4 Hz to 73.4 Hz, corresponding to a rise in pitch from C to D?

15.72 • Holding Up Under Stress. A string or rope will break apart if it is placed under too much tensile stress [see Eq. (11.8)]. Thicker ropes can withstand more tension without breaking because the thicker the rope, the greater the cross-sectional area and the smaller the stress. One type of steel has density 7800 kg/m^3 and will break if the tensile stress exceeds $7.0 \times 10^8 \text{ N/m}^2$. You want to make a guitar string from 4.0 g of this type of steel. In use, the guitar string must be able to withstand a tension of 900 N without breaking. Your job is to determine (a) the maximum length and minimum radius the string can have; (b) the highest possible fundamental frequency of standing waves on this string, if the entire length of the string is free to vibrate.

15.73 •• DATA In your physics lab, an oscillator is attached to one end of a horizontal string. The other end of the string passes over a frictionless pulley. You suspend a mass *M* from the free end of the string, producing tension *Mg* in the string. The oscillator produces transverse waves of frequency *f* on the string. You don't vary this frequency during the experiment, but you try strings with three different linear mass densities μ . You also keep a fixed distance between the end of the string where the oscillator is attached and the point where the string is in contact with the pulley's rim. To produce standing waves on the string, you vary *M*; then you measure the node-to-node distance *d* for each standing-wave pattern and obtain the following data:

String	A	A	B	B	C
μ (g/cm)	0.0260	0.0260	0.0374	0.0374	0.0482
M (g)	559	249	365	207	262
d (cm)	48.1	31.9	32.0	24.2	23.8

(a) Explain why you obtain only certain values of d . (b) Graph μd^2 (in $\text{kg} \cdot \text{m}$) versus M (in kg). Explain why the data plotted this way should fall close to a straight line. (c) Use the slope of the best straight-line fit to the data to determine the frequency f of the waves produced on the string by the oscillator. Take $g = 9.80 \text{ m/s}^2$. (d) For string A ($\mu = 0.0260 \text{ g/cm}$), what value of M (in grams) would be required to produce a standing wave with a node-to-node distance of 24.0 cm? Use the value of f that you calculated in part (c).

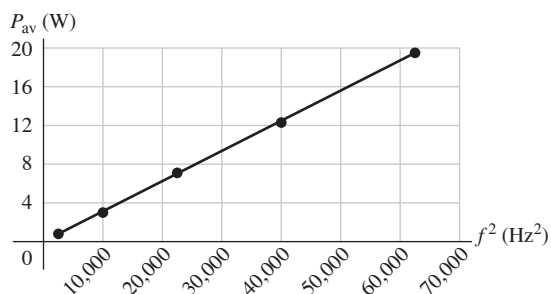
15.74 •• DATA Scale length is the length of the part of a guitar string that is free to vibrate. A standard value of scale length for an acoustic guitar is 25.5 in. The frequency of the fundamental standing wave on a string is determined by the string's scale length, tension, and linear mass density. The standard frequencies f to which the strings of a six-string guitar are tuned are given in the table:

String	E2	A2	D3	G3	B3	E4
f (Hz)	82.4	110.0	146.8	196.0	246.9	329.6

Assume that a typical value of the tension of a guitar string is 78.0 N (although tension varies somewhat for different strings). (a) Calculate the linear mass density μ (in g/cm) for the E2, G3, and E4 strings. (b) Just before your band is going to perform, your G3 string breaks. The only replacement string you have is an E2. If your strings have the linear mass densities calculated in part (a), what must be the tension in the replacement string to bring its fundamental frequency to the G3 value of 196.0 Hz?

15.75 •• DATA You are measuring the frequency dependence of the average power P_{av} transmitted by traveling waves on a wire. In your experiment you use a wire with linear mass density 3.5 g/m. For a transverse wave on the wire with amplitude 4.0 mm, you measure P_{av} (in watts) as a function of the frequency f of the wave (in Hz). You have chosen to plot P_{av} as a function of f^2 (Fig. P15.75). (a) Explain why values of P_{av} plotted versus f^2 should be well fit by a straight line. (b) Use the slope of the straight-line fit to the data shown in Fig. P15.75 to calculate the speed of the waves. (c) What angular frequency ω would result in $P_{\text{av}} = 10.0 \text{ W}$?

Figure P15.75



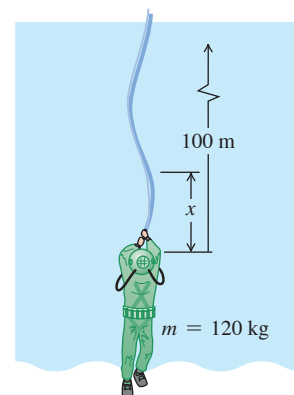
CHALLENGE PROBLEMS

15.76 •• A rectangular neoprene sheet has width $W = 1.00 \text{ m}$ and length $L = 4.00 \text{ m}$. The two shorter edges are affixed to rigid steel bars that are used to stretch the sheet taut and horizontal. The force applied to either end of the sheet is $F = 81.0 \text{ N}$. The sheet has a total mass $M = 4.00 \text{ kg}$. The left edge of the sheet is wiggled vertically in a uniform sinusoidal motion with amplitude $A = 10.0 \text{ cm}$ and frequency

$f = 1.00 \text{ Hz}$. This sends waves spanning the width of the sheet rippling from left to right. The right side of the sheet moves upward and downward freely as these waves complete their traversal. (a) Use a two-dimensional generalization of the discussion in Section 15.4 to derive an expression for the velocity with which the waves move along the sheet in terms of generic values of W, L, F, M, f , and A . What is the value of this speed for the specified choices of these parameters? (b) If the positive x -axis is oriented rightward and the steel bars are parallel to the y -axis, the height of the sheet may be characterized as $z(x, y) = A \sin(kx - \omega t)$. What is the value of the wave number k ? (c) Write down an expression with generic parameters for the rate of rightward energy transfer by the slice of sheet at a given value of x at generic time t . (d) The power at $x = 0$ is supplied by the agent wiggling the left bar upward and downward. How much energy is supplied each second by that agent? Express your answer in terms of generic parameters and also as a specific energy for the given parameters.

15.77 ••• CP CALC A deep-sea diver is suspended beneath the surface of Loch Ness by a 100-m-long cable that is attached to a boat on the surface (Fig. P15.77). The diver and his suit have a total mass of 120 kg and a volume of 0.0800 m^3 . The cable has a diameter of 2.00 cm and a linear mass density of $\mu = 1.10 \text{ kg/m}$. The diver thinks he sees something moving in the murky depths and jerks the end of the cable back and forth to send transverse waves up the cable as a signal to his companions in the boat. (a) What is the tension in the cable at its lower end, where it is attached to the diver?

Figure P15.77



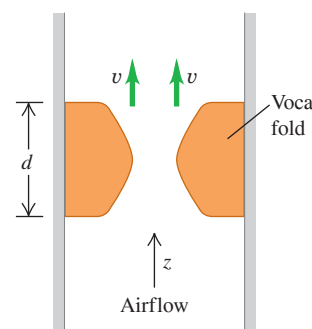
Do not forget to include the buoyant force that the water (density 1000 kg/m^3) exerts on him. (b) Calculate the tension in the cable a distance x above the diver. In your calculation, include the buoyant force on the cable. (c) The speed of transverse waves on the cable is given by $v = \sqrt{F/\mu}$ [Eq. (15.14)]. The speed therefore varies along the cable, since the tension is not constant. (This expression ignores the damping force that the water exerts on the moving cable.) Integrate to find the time required for the first signal to reach the surface.

MCAT-STYLE PASSAGE PROBLEMS

BIO Waves on Vocal Folds. In the larynx, sound is produced by the vibration of the *vocal folds* (also called “vocal cords”).

Figure P15.79

The accompanying figure is a cross section of the vocal tract at one instant in time. Air flows upward (in the $+z$ -direction) through the vocal tract, causing a transverse wave to propagate vertically upward along the surface of the vocal folds. In a typical adult male, the thickness of the vocal folds in the direction of airflow is $d = 2.0 \text{ mm}$. High-speed photography shows that for a frequency of vibration of $f = 125 \text{ Hz}$, the wave along the surface of the vocal folds travels upward at a speed of $v = 375 \text{ cm/s}$. Use t for time, z for displacement in the $+z$ -direction, and λ for wavelength.

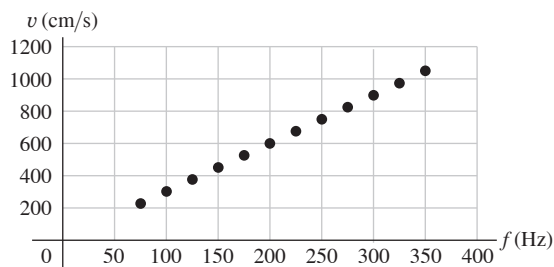


15.78 What is the wavelength of the wave that travels on the surface of the vocal folds when they are vibrating at frequency f ? (a) 2.0 mm; (b) 3.3 mm; (c) 0.50 cm; (d) 3.0 cm.

15.79 Which of these is a possible mathematical description of the wave in Problem 15.78? (a) $A \sin[2\pi f(t + z/v)]$; (b) $A \sin[2\pi f(t - z/v)]$; (c) $A \sin(2\pi ft) \cos(2\pi z/\lambda)$; (d) $A \sin(2\pi ft) \sin(2\pi z/\lambda)$.

15.80 The wave speed is measured for different vibration frequencies. A graph of the wave speed as a function of frequency (Fig. P15.80) indicates that as the frequency increases, the wavelength (a) increases; (b) decreases; (c) doesn't change; (d) becomes undefined.

Figure P15.80



ANSWERS

Chapter Opening Question ?

(iii) The power of a mechanical wave depends on both its amplitude and its frequency [see Eq. (15.25)].

Key Example VARIATION Problems

VP15.3.1 (a) 0.196 Hz (b) 5.98 m/s (c) 1.23 rad/s (d) 0.206 m^{-1}

VP15.3.2 (a) $T = 8.00 \times 10^{-3} \text{ s}$, $\lambda = 1.96 \text{ m}$

(b) $f = 81.7 \text{ Hz}$, $\omega = 513 \text{ rad/s}$

VP15.3.3 (a) 111 m/s (b) 4.44 m

(c) $y(x = +2.50 \text{ m}, t) = (5.00 \text{ mm}) \cos [(3.54 \text{ rad}) - (157 \text{ rad/s})t]$

VP15.3.4 (a) 36.0 m/s (b) 0.0193 kg/m

VP15.5.1 (a) 10.9 rad/s (b) 0.294 W

VP15.5.2 $2.50 \times 10^{-3} \text{ m}$

VP15.5.3 (a) 0.159 W/m^2 (b) 3.76 m

VP15.5.4 $6.94 \times 10^{-12} \text{ J}$

VP15.8.1 (a) 64.0 m/s (b) 2.25 m/s (c) $3.62 \times 10^3 \text{ m/s}^2$

VP15.8.2 (a) 251 m/s (b) 145 N

VP15.8.3 (a) 1.20 m (b) 80.0 Hz (c) $1.90 \times 10^{-2} \text{ kg/m}$

VP15.8.4 (a) 76.7 N (b) 1.34 m

Bridging Problem

$$(a) \quad F(r) = \frac{m\omega^2}{2L}(L^2 - r^2)$$

$$(b) \quad v(r) = \omega \sqrt{\frac{L^2 - r^2}{2}}$$

$$(c) \quad \frac{\pi}{\omega\sqrt{2}}$$